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What is the best age to circumcise?

A medical and ethical analysis

by Alex Myers and Brian D. Earp

Abstract

Penile circumcision is often claimed to be simpler, safer, and more cost-effective when performed in the neonatal period as opposed to later in life, with a greater benefit-to-risk ratio. In the first part of this paper, we critically examine the evidence base for these claims, and find that it is not as robust as is commonly assumed. In the second part, we demonstrate that, even if one simply grants these claims for the sake of argument, it still does not follow that neonatal circumcision is ethically permissible - absent urgent medical necessity. Based on a careful consideration of the relevant evidence, arguments, and counterarguments, we conclude that medically unnecessary penile circumcision—like other medically unnecessary genital procedures, such as 'cosmetic' labiaplasty—should not be performed on an individual who is too young (or otherwise unable) to provide meaningful consent to the procedure.

Key words: circumcision, benefits, risks, autonomy, consent, gender

1. Introduction

Over recent decades, two major trends have emerged in the academic literature concerning penile circumcision (partial or total removal of the penile prepuce). On the one hand, findings from three randomized controlled trials (RCTs) conducted in Africa in the 2000s[1] have been taken to show that clinical circumcision of adult males can reduce their risk of contracting HIV through penile-vaginal intercourse, at least in settings with a high prevalence of heterosexual HIV transmission and a low prevalence of such circumcision. Although the quality of the evidence from these trials has been called into question,[2] with additional concerns raised about the decision pathways by which this evidence was rushed into policy,[3] and reports of an apparent lack of real-world effectiveness at the population level now beginning to emerge,[4] a heavily-funded campaign to

[1] Auvert, B., Taljaard, D., Lagarde, E., et al. (2005). Randomized, controlled intervention trial of male circumcision for reduction of HIV infection risk: the ANRS 1265 Trial. *PLoS Med,* 2(11*)*, e298; Bailey, R. C., Moses, S., Parker, C. B., et al. (2007). Male circumcision for HIV prevention in young men in Kisumu, Kenya: a randomised controlled trial. *Lancet,* 369(9562), 643–656; Gray, R.H., Kigozi, G., Serwadda, D., et al. (2007). Male circumcision for HIV prevention in men in Rakai, Uganda: a randomised trial. *Lancet,* 369(9562), 657–666. A fourth trial looking at male to female transmission of HIV as a consequence of penile circumcision was stopped early due to "futility" with female partners of circumcised men becoming infected with HIV at a higher rate than female partners of genitally intact men; see Wawer, M. J., Makumbi, K., Kigozi, G., et al. (2009). Circumcision in HIV-infected men and its effect on HIV transmission to female partners in Rakai, Uganda: a randomised controlled trial. *Lancet,* 374(9685): 229-237.

[2] Van Howe, R. S., & Boyle, G. (2018). Meta-analysis of HIV acquisition studies incomplete and unstable. *BJUI International*, e-letter. Available at http://www.bjuinternational.com/letters/meta-analysis-circumcision-incomplete-unstable/ (accessed November 3, 2019); Boyle, G. J., & Hill, G. (2011). Sub-Saharan African randomised clinical trials into male circumcision and HIV transmission: methodological, ethical and legal concerns. *Journal of Law and Medicine,* 19(2), 316–334; Van Howe, R. S., & Storms, M. R. (2011). How the circumcision solution in Africa will increase HIV infections. *Journal of Public Health in Africa,* 2(1), 11–15; Garenne, M., Giami, A., & Perrey, C. (2013). Male circumcision and HIV control in Africa: questioning scientific evidence and the decision-making process. In T. Giles-Vernik & J. Webb (Eds.), *Global health in Africa: historical perspectives on disease control* (pp. 185-210). Athens: Ohio University Press. For a contrary view, see, e.g., Wamai, R. G. et al. (2012) "Criticisms of African trials fail to withstand scrutiny: male circumcision does prevent HIV infection," *Journal of Law and Medicine 20*(1), 93–123.

[3] Dowsett, G. W., & Couch, M. (2007). "Male circumcision and HIV prevention: is there really enough of the right kind of evidence?" *Reproductive Health Matters*, 15(29): 33–44; Bell, K. (2015). HIV prevention: making male circumcision the 'right' tool for the job. *Global Public Health,* 10(5–6): 552–572; de Camargo, K. R. Jr., de Oliveira Mondonça, A. L., Perrey, C., & Giami, A. (2013). Male circumcision and HIV: a controversy study on 'facts' and 'values. *Global Public Health,* 8(7), 769–783; de Camargo, K. R., Jr., de Oliveira Mondonça, A. L., Perrey, C., & Giami, A. (2015). Making the circumcision controversy controversial: going meta and taking aim at the messenger(s): reply to Wamai et al. *Global Public Health,* 10(5–6), 667–671. The dominant policy paradigm appears to have been driven by a relatively small network of actors with influence at the WHO among other organizations; see Giami A., Perrey, C., de Oliveria Mendonça, A.L., & de Camargo, K. R. (2015). Hybrid forum or network? The social and political construction of an international 'technical consultation': male circumcision and HIV prevention. *Global Public Health,* 10(5–6), 589–606.

[4] Or even higher rates of infection among some medically circumcised males: Rosenberg, M. S., Gómez-Olivé, F. X., Rohr, J. K., Kahn, K., & Bärnighausen, T. W. (2018). Are circumcised men safer sex partners? Findings from the HAALSI cohort in rural South Africa. PloS one, 13(8), e0201445. See also Garenne, M., & Matthews, A. (2019). Voluntary medical male circumcision and HIV in Zambia: expectations and observations. *Journal of*

"scale up" mass circumcision of African boys and men as a public health measure is currently underway.[5]

On the other hand, human rights scholars have increasingly objected to the medically unnecessary genital cutting of any person—whether female, male, or intersex—who either does not or cannot consent to the cutting. Such objections were recently summarized in an international consensus statement published in the *American Journal of Bioethics*, in which the authors argued that under most conditions, "cutting any person's genitals without their informed consent is a serious violation of their right to bodily integrity. As such, it is impermissible unless the person is non-autonomous (incapable of consent) and the cutting is medically necessary" (see Box 1) and thus "cannot reasonably be deferred."[6]

Biosocial Science, online ahead of print at https://doi.org/10.1017/S0021932019000634. However, see Davis, S., Toledo, C., Lewis, L., Maughan-Brown, B., Ayalew, K., & Kharsany, A. B. (2019). Does voluntary medical male circumcision protect against sexually transmitted infections among men and women in real-world scale-up settings? Findings of a household survey in KwaZulu-Natal, South Africa. BMJ Global Health, 4(3), e001389. For preliminary evidence of risk-compensation or behavioral disinhibition as a potential mechanism for increased STI acquisition in some populations over the long term—based on biomarkers and a randomized design—see Kim, H. B., Pop-Eleches, C., Jung, J., & Kim, B. (2018). Male circumcision, peer effects and risk compensation. Presented to the Association of Public Policy Analysis and Management, November 10, 2018. Available at http://barrett.dyson.cornell.edu/NEUDC/paper_226.pdf (accessed November 3, 2019).

[5] Hankins, C., Forsythe, S., & Njeuhmeli, E. (2011). Voluntary medical male circumcision: an introduction to the cost, impact, and challenges of accelerated scaling up. *PLoS Med*, 8(11): e1001127; Torres-Rueda, S., Wambura, M., Weiss, H., et al. (2018). Cost and cost-effectiveness of a demand creation intervention to increase uptake of voluntary medical male circumcision in Tanzania: spending more to spend less. *JAIDS: Journal of Acquired Immune Deficiency Syndromes,* 78(3*)*, 291-299. Originally termed "Voluntary Medical Male Circumcision" (VMMC) with the aim of recruiting men to be circumcised voluntarily, the scale-up now appears to be targeting younger and younger males, including until recently infants by way of their mothers, who are presumably much less capable (or incapable) of refusing the procedure. See, for example, Sgaier, S. K., Sharma, S., Eletskaya, M., et al. (2017). Attitudes and decision-making about early-infant versus early-adolescent male circumcision: demand-side insights for sustainable HIV prevention strategies in Zambia and Zimbabwe. *PLoS ONE,* 12(7), e0181411; Sidler, D., Earp, B. D., van Niekerk, A. A., Moodley, K., & Kling, S. (2017). Targeting mothers and selling men what they do not want: a response to 'missed opportunities for circumcision of boys.' *South African Medical Journal, 107(4)*, 281. There are now disturbing indications that many thousands of young boys have been forcibly circumcised without parental knowledge or consent as a part of this scale-up: Luseno, W. K., Field, S. H., Iritani, B. J., et al. (2019). Consent challenges and psychosocial distress in the scale-up of Voluntary Medical Male Circumcision among adolescents in Western Kenya. *AIDS and Behavior*, online ahead of print at https://link.springer.com/article/10.1007/s10461-019-02620-7. Further problems include "potentially misleading or questionable mobilization practices (including possibly undue inducements), problematic uses of social pressure, and circumcision of children under age ten." Gilbertson et al. (2019). Voluntary medical male circumcision for HIV prevention among adolescents in Kenya: unintended consequences of pursuing service delivery targets. *PLOS ONE*, 14(11): e0224548. See also the 2019 report of the VMMC Experience Project, an African-led NGO, alleging widespread human rights abuses associated with the campaign, as submitted to the United Nations Committee on the Rights of the Child: https://www.vmmcproject.org/un-report/ (accessed November 3, 2019).

[6] The Brussels Collaboration on Bodily Integrity (BCBI). (2019). Medically unnecessary genital cutting and the rights of the child: moving toward consensus. *American Journal of Bioethics*, 19(10), 17-28. The authors clarify that the scope of their argument for purposes of the statement is the so-called Western medicolegal context, especially countries with a strong tradition of individual rights. For arguments and evidence in support of the main thesis of the "Brussels" statement that go into greater depth, see Lightfoot-Klein, H., Chase, C., Hammond, T., & Goldman, R. (2000). Genital surgeries on children below an age of consent. In L. T. Szuchman & F. Muscarella (Eds.), in *Psychological perspectives on human sexuality* (pp. 440-479). New York: John Wiley & Sons; Svoboda,

Building on this perspective, we argue that the best age to circumcise, considering both empirical/medical and normative/ethical factors, is no earlier than an appropriate age of consent (depending on the jurisdiction, the age of consent to sexual relations, for instance, is usually between 14 and 18).[7] At that age, the individual's own informed consent can at least potentially be obtained, and indeed must be obtained for the circumcision to be morally permissible.[8]

Our proposal is motivated in part by a concern for children's sexual self-determination. In modern Western societies, children of all sexes and genders are taught from an early age that adults should not touch—let alone cut or remove healthy tissue from—their so-called "private parts" before they are sexually autonomous. The only widely recognized exceptions to this rule concern (1) the youngest of children, insofar as they are not yet able to wash themselves adequately and need help with this from a designated caretaker and (2) required medical examinations or procedures. The emphasis here is on "required." Under ordinary conditions, if a doctor or other healthcare provider handles a child's genitals beyond what is "strictly necessary for diagnosis or treatment," this is immediately understood to cross an ethical line. Moreover, if the child-patient happens to be unconscious or otherwise does not remember the medically unnecessary genital interference, this does not normally make the action morally permissible.[9]

J. S. (2013). Promoting genital autonomy by exploring commonalities between male, female, intersex, and cosmetic female genital cutting. *Global Discourse,* 3(2), 237–255; Townsend, K. G. (2019). The child's right to genital integrity. *Philosophy & Social Criticism*, online ahead of print at https://doi.org/10.1177/0191453719854212; Kehrer, I. (2019). Cuts into children's future: a comparative analysis between FGM, male circumcision and intersex genital surgeries. *Peace Human Rights Governance*, 3(3), 333-363. See also Earp, B. D., & Steinfeld, R. (2017). Gender and genital cutting: a new paradigm. In T. G. Barbat (Ed.), *Gifted women, fragile men*. Euromind Monographs 2. Brussels: ALDE Group-EU Parliament; Earp, B. D., & Steinfeld, R. (2018). Genital autonomy and sexual well-being. *Current Sexual Health Reports,* 10, 7–17.

[7] See https://www.ageofconsent.net/world (accessed November 3, 2019). Please note that although most jurisdictions have a set age at which capacity to consent to sexual relations is presumed for legal purposes, the moral capacity to provide valid consent (whether to sex or other activities) in adolescence develops variably and individually. Some adolescents who have reached the age at which they can legally consent to sex, for example, might nevertheless not be Gillick competent for purposes of deciding about certain medical procedures (and vice versa). On Gillick competence, see Larcher, V., & Hutchinson, A. (2010). How should paediatricians assess Gillick competence? *Archives of Disease in Childhood*, 95(4), 307-311.

[8] For an overview of debates concerning the capacity of legal minors to consent to certain medical treatments see for example Alderson, P. (2007). Competent children? Minors' consent to health care treatment and research. *Social Science & Medicine*, 65(11), 2272-2283. Note, however, that medically unnecessary circumcision does not normally constitute treatment (on most common understandings of that term).

[9] Both quotations from BCBI, "Medically unnecessary genital cutting," *op cit.* note 6. For related arguments, see Myers, A. (2015). Neonatal male circumcision, if not already commonplace, would be plainly unacceptable by modern ethical standards. *The American Journal of Bioethics,* 15(2): 54–55.

Box 1. Medically necessary versus medically beneficial[10]

According to the Brussels Collaboration on Bodily Integrity (BCBI), an intervention to alter a bodily state is medically necessary when (1) the bodily state poses a serious, time-sensitive threat to the person's well-being, typically due to a functional impairment in an associated somatic process, and (2) the intervention, as performed without delay, is the least harmful feasible means of changing the bodily state to one that alleviates the threat. "Medically necessary" is therefore different from "medically beneficial," a weaker standard, which requires only that the expected health-related benefits outweigh the expected health-related harms.[11] The BCBI states that although the weaker, "medically beneficial" standard may well be appropriate for certain interventions into the body, it is not appropriate for cutting or removing healthy tissue from the genitals of a non-consenting person (given, among other things, the special significance of the body part in question).

An opposing view has been advanced by those who advocate for non-voluntary neonatal male circumcision (NNMC).[12] According to these advocates (Box 2), penile circumcisions performed in infancy (roughly, the first year of life) and especially in the neonatal period (roughly, the first month of life) are relevantly different from circumcisions performed later in life. Specifically, they claim that penile circumcision is technically simpler, safer, and more cost effective when it is performed on an infant or neonate as opposed to an adolescent or adult, and that the accumulated health benefits are also greater.[13] Thus, to delay penile circumcision until it

[10] This Box is adapted from Box 1 of BCBI, *op cit*. note 6. Internal references omitted. The definition of medical necessity is based on Earp, B. D. (2019). The child's right to bodily integrity. In D. Edmonds (Ed.), *Ethics and the contemporary world* (pp. 217-235). Abingdon and New York: Routledge.

[11] The BCBI notes that benefit/harm comparisons are often contested as they depend on "the specific weights assigned to the possible outcomes of the intervention, given among other things a) the subjective value to the individual of the body parts that may be affected, (b) the individual's tolerance for different kinds or degrees of risk to which those body parts may be exposed, and (c) any preferences the individual may have for alternative (e.g., less risky or invasive) means of pursuing the intended health-related benefits." BCBI, *op cit*. note 6, p. 18. For in-depth discussions of the contested nature of benefit/harm ratios concerning male circumcision in particular, emphasizing that evaluations of the primary literature may be shaped by cultural bias or other potentially distorting factors, see Darby, R. (2015). Risks, benefits, complications and harms: neglected factors in the current debate on non-therapeutic circumcision. *Kennedy Institute of Ethics Journal,* 25(1), 1–34; Earp, B. D., & Shaw, D. M. (2017). Cultural bias in American medicine: the case of infant male circumcision. *Journal of Pediatric Ethics* 1(1), 8–26; Frisch, M., Aigrain, Y., Barauskas, Y., et al. (2013). Cultural bias in the AAP's 2012 technical report and policy statement on male circumcision. *Pediatrics,* 131(4), 796–800. See also Van Howe, R. S. (2018). Response to Vogelstein: How the 2012 AAP Task Force on circumcision went wrong. *Bioethics*, 32(1), 77-80; and Svoboda, J. S., & Van Howe, R. S. (2013). Out of step: fatal flaws in the latest AAP policy report on neonatal circumcision. *Journal of Medical Ethics*, 39(7), 434-441 (replies and counter-replies are available at the journal website).

[12] In keeping with a familiar distinction in bioethics, we use the term "non-voluntary," which concerns a moral patient unable to express their will in relation to a given procedure, rather than "involuntary" which implies a moral agent able to express their will who refuses the procedure.

[13] See for instance Dickson, K. E., Samuelson, J., Ashengo, T. A., Chrouser, K., Curran, K., Otolorin, E., Caldamone, A., & Tomlinson, D. (2010). Manual for early infant male circumcision under local anesthesia. World Health Organization and Jhpeigo. Geneva: WHO Press. According to the *Los Angeles Times*, one of the main

could be undertaken voluntarily, they claim, is to increase the associated risks and costs (both medical and monetary), as well as to reduce the associated benefits, *assuming the procedure will be performed either way.* Whether this latter assumption is an appropriate one will be discussed later. Nevertheless, for at least some advocates of NNMC, such financial and health-related considerations are so compelling that they may override the default moral presumption against medically unnecessary interference with a child's genitals—not only by touching in this case, but indeed by cutting into and irreversibly excising healthy and functional tissue.[14]

Against this moral presumption, it is often noted that pre-autonomous children are not yet capable of consenting to any surgical procedure, medically necessary or otherwise. Hence, it is claimed, the child's parents must decide about such matters. It is true that parents are normally (and appropriately) entrusted with many decisions surrounding the care and upbringing of their children. But in contemporary Western societies, at least, parents are not permitted to authorize simply whatever encroachments into their child's bodily integrity they may choose. Rather, there are clear moral and legal limitations. For example, in addition to encroachments involving genital contact of any kind, those that involve cutting into the skin, acute tissue damage, or permanent modification of the external physical features of a child, must normally clear a very high bar to be considered ethically acceptable. In fact, some encroachments, such as child facial scarification, foot binding, and medically unnecessary female genital cutting—including "symbolic" forms that are less invasive than penile circumcision—are categorically forbidden in such societies.[15] Even

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authors of this manual, David Tomlinson, was then serving as "the World Health Organization's chief circumcision expert." Hennessy-Fiske, M. (2011, September 2). Injuries linked to circumcision clamps. *Los Angeles Times.* Available at https://www.latimes.com/health/la-xpm-2011-sep-26-la-he-circumcision-20110926-story.html (accessed November 3, 2019). Prior to consulting for the WHO, it appears that Tomlinson had filed for a patent for a newborn circumcision clamp, claimed to be preferable to those most widely in use: https://patents.justia.com/patent/20080004631 (accessed November 3, 2019). Notably, Tomlinson's clamp is touted in the WHO manual for potential mass adoption in Africa "for the purposes of HIV prevention," despite no empirical evidence that newborn circumcision specifically has a protective effect against HIV. Instead, the manual repeatedly conflates adult and newborn circumcision, citing evidence drawn from studies of the former to justify claims about the latter. A similar conflation problem can be found in Morris, B. J. et al. (2012). A 'snip' in time: What is the best age to circumcise? *BMC Pediatrics*, 12(20), 1-15. It should be noted that these authors appear to assume that the circumcision will be performed at some point or another. However, if an adult decides stay genitally intact, as most do when given the opportunity, the hypothetical circumcision will be even "simpler, safer and more cost-effective" because it will not occur.

[14] That is, the penile prepuce. Cold, C. J., & Taylor, J. R. (1999). The prepuce. *BJU International,* 83(S1), 34–44.

[15] One such "symbolic" form is so-called ritual nicking of the clitoral hood/foreskin, which does not remove tissue and often leaves no visible mark on the vulva. According to a recent ruling from Australia's highest court, for legal purposes, (1) the clitoral foreskin is part of the clitoris, and (2) any medically unnecessary cutting of this tissue, no matter how slight, constitutes illegal mutilation, irrespective of whether there is any lasting damage or functional impairment. Australian Associated Press. (2019, October 16). High Court upholds NSW genital mutilation convictions. *Sydney Morning Herald*. Available at https://www.smh.com.au/national/nsw/high-court-upholds-nsw-genital-mutilation-convictions-20191016-p53197.html (accessed November 3, 2019). For further discussion of ritual nicking and different Western legal standards for male versus female genital cutting, see Davis, D. S. (2001).

slapping or spanking a child for their own supposed good (i.e., as a means of discipline) is banned in more than 50 countries.[16]

When a child faces a medical emergency such that a bodily encroachment cannot be delayed without significantly increasing the risk of death or disability, this creates a special situation. In such an emergency situation, the permission of the child's parents or other appropriate caregiver must be relied upon as the closest available proxy for the child's consent, assuming the child is not yet in a position to consent on their own behalf.[17] By contrast, when there is no disease or deformity present for which immediate medical intervention is required, it is normally understood that the permission for a surgery must come from the individual whose body will be altered and/or exposed to the associated risk. As Hutson argues:

> The most fundamental principle of surgery is that no operation should be done if there is no disease, as it cannot be justified if the risk of the procedure is not balanced by the risk of a disease. Even when patients have significant disease, potentially dangerous operations can hardly be justified if their risks are much greater than the disease itself. The problem for routine circumcision is that since there is no disease, no complication whatsoever can be tolerated, since the risks of the procedure are not being balanced against the risks of any present disease.[18]

Against this view, NNMC advocates (Box 2) argue that *future* diseases must also be considered; that is, potential future diseases to which the individual might be exposed, depending largely on their own choices, and for which there are both less risky and more effective modes of prevention that do not involve genital cutting. We address this argument in the present paper. In doing so, we carefully examine some of the main empirical and conceptual claims put forward by advocates of NNMC. We argue, first, drawing on recent evidence, that the strictly medical case for NNMC is not as compelling as its advocates maintain. But going beyond this, we suggest that the narrow parameters of a medical cost-benefit analysis are insufficient to draw meaningful *moral* conclusions about the acceptability of NNMC.[19] Instead, what is required is more

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Male and female genital alteration: a collision course with the law. *Health Matrix,* 11(1), 487–570; Davis, D. S. (2003). Cultural bias in responses to male and female genital surgeries. *The American Journal of Bioethics,* 3(2), W15-W16; Earp, B. D., Hendry, J., & Thomson, M. (2017). Reason and paradox in medical and family law: shaping children's bodies. *Medical Law Review*, 25(4), 604-627; and Earp, B. D. (2020). Protecting children from medically unnecessary genital cutting without stigmatizing women's bodies: implications for sexual pleasure and pain. *Archives of Sexual Behavior*, available online ahead of print at https://doi.org/10.1007/s10508-020-01633-x.

[16] Cole, D. (2019, September 3). Whatever happened to the campaign to ban … spanking? *NPR*. Available at https://www.npr.org/sections/goatsandsoda/2019/09/03/755136377/whatever-happened-to-the-campaign-to-ban-spanking (accessed November 3, 2019).

[17] See Alderson, *op cit*. note 8.

[18] Hutson, J. M. (2004). Circumcision: a surgeon's perspective. *Journal of Medical Ethics,* 30(3), 238–240.

[19] See also Darby, *op cit*. note 11.

thoroughgoing engagement with the particular ethical issues posed by medically unnecessary genital cutting of minor children.

Box 2. Who advocates for non-voluntary neonatal male circumcision (NNMC)?

Throughout this article, we refer to the arguments of NNMC advocates, who have been described as falling into two main camps: (1) strong advocates, who argue that the health benefits of NNMC are so great and the risks so trivial that boys should be "circumcised as a matter of routine, or even that the operation should be compulsory," and (2) weak advocates, who believe that "while the benefits outweigh the risks they are not so great that doctors should recommend the operation, but great enough to authorize parental discretion."[20] The "strong" position is associated primarily with a grouping around Brian J. Morris, a former professor of molecular medical sciences at the University of Sydney and a driving force behind the Circumcision Academy of Australia (CAA), a network of circumcision proponents and financially-interested providers.[21] The "weak" position is associated primarily with certain actors within the US medical establishment, including, most notably, the eight-member Task Force behind the now-expired 2012 circumcision policy statement of the American Academy of Pediatrics (AAP).[22] Rejecting both positions—that is, failing to conclude that the benefits of NNMC outweigh the risks, or reaching the opposite conclusion, that the risks outweigh the benefits—are most child health authorities outside the United States, including all comparable national-level medical bodies to have issued formal statements on NNMC (see later discussion).

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[20] These quotes are from Darby, *op cit*. note 11, pp. 1-2, summarizing the position of NNMC advocates. Some of the material in this Box is lightly adapted from Darby's discussion.

[21] See https://www.circumcisionaustralia.org (accessed November 3, 2019). The CAA lobbies for government health insurance coverage for medically unnecessary newborn circumcisions; billing for such circumcisions appears to be a primary source of income for several CAA board members, including some whose private practices are centered around offering the procedure. For details, see Frisch, M., & Earp, B. D. (2018). Circumcision of male infants and children as a public health measure in developed countries: a critical assessment of recent evidence. *Global Public Health,* 13(5), 626–641, especially note 4.

[22] AAP Task Force on Circumcision. (2012). Circumcision policy statement. *Pediatrics*, 130(3), 585-586. Although the AAP policy formally expired in 2017, an anonymous working group at the US Centers for Disease Control and Prevention (CDC) subsequently released a similar policy, in 2018, which remains online: https://www.cdc.gov/hiv/risk/male-circumcision.html (accessed November 3, 2019). This release came in the face of extensive criticism from international medical authorities and at least one invited peer reviewer, many of whose substantive objections to an earlier draft of the CDC policy were effectively ignored during the process of revision. See Kupferschmid, C., et al. (2015). Commentary on the CDC "Recommendations for providers counseling male patients and parents regarding male circumcision and the prevention of HIV infection, STIs, and other health outcomes." CDC Public Submission. Available at https://www.regulations.gov/document?D=CDC-2014-0012-2455 (accessed November 3, 2019); and Van Howe, R. S. (2015). A CDC-requested, evidence-based critique of the Centers for Disease Control and Prevention 2014 draft on male circumcision: How ideology and selective science lead to superficial, culturally-biased recommendations by the CDC. CDC Peer Review Report. DOI: 10.13140/2.1.1148.4964. For additional articulations of the "weak" position on NNMC as described above, see Benatar, M., & Benatar, D. (2003). Between prophylaxis and child abuse: the ethics of neonatal male circumcision. *American Journal of Bioethics,* 3(2), 35-48; Mazor, J. (2013). The child's interests and the case for the permissibility of male infant circumcision. *Journal of Medical Ethics,* 39(7), 421–428; and Jacobs, A. J., & Arora, K. S. (2015). Ritual male infant circumcision and human rights. *The American Journal of Bioethics*, 15(2), 30-39.

In the end, we argue that children's interest in future bodily autonomy, especially over those parts of the body that have special psychosexual significance—such as the penis, vulva, or breasts—is sufficiently weighty to render NNMC morally impermissible in most cases.[23] Even taking health considerations into account, that is, we find that the best age to circumcise is the age at which the individual is developmentally capable of providing informed consent to the permanent alteration of their own genitals.

As a final note, we will be using gender-neutral language (as in the previous sentence) as far as possible in this paper while staying true to particular empirical findings. This is because we recognize that some people with penises may not identify as boys or men, such as trans women or some genderqueer individuals.[24] In fact, the potential harms of neonatal or early-childhood circumcision for trans women who elect a penile inversion surgery—as a part of gender-affirming care, for example—has yet to receive much attention.[25] However, the preemptive removal of a large proportion of sensitive, elastic genital tissue from the penis that could otherwise have been used in the construction of a neovagina—i.e., the penile foreskin—is undoubtedly of relevance to the welfare interests of such women.[26]

2. Empirical issues

A. Benefits

Let us begin by looking at the issue of health benefits, a number of which have been claimed to follow from penile circumcision, although their likelihood, magnitude, and causal relationship to

[23] See the discussion by Munzer, who refers to the physical and symbolic "salience" of the genitals as a relevant moral consideration. Munzer, S. R. (2018). Examining nontherapeutic circumcision. *Health Matrix*, 28(1), 1-78.
[24] See, e.g., Dembroff, R. (2019). Beyond binary: genderqueer as critical gender kind. *Philosopher's Imprint*, online ahead of print at http://philsci-archive.pitt.edu/16317/; Vogler, S. (2019). Determining transgender: adjudicating gender identity in US asylum law. *Gender & Society*, 33(3), 439-462.
[25] As the authors of a recent review state: "The inner layer of the foreskin, the neurovascular bundle, and the glans are used as a pedicled island flap. This flap is used to construct the inner part of the labia minora, the neoclitoris, and its prepuce." They note that in cases where "more tissue might be required (e.g., because of circumcision or a small penis), further full-thickness skin grafts (e.g., from the proximal medial thigh)" will likely have to be transplanted. Papadopulos, N. A., Lellé, J. D., Zavlin, D., Herschbach, P., Henrich, G., Kovacs, L., ... & Schaff, J. (2017). Quality of life and patient satisfaction following male-to-female sex reassignment surgery. *The Journal of Sexual Medicine*, 14(5), 721-730.
[26] For an intriguing discussion of certain tensions in this area, see Levy, I. (2017, June 23). Transgender and Jewish. *Tablet Magazine*. Available at https://www.tabletmag.com/jewish-life-and-religion/237080/transgender-and-jewish (accessed November 3, 2019).

NNMC in particular remain in dispute.[27] This dispute can be illustrated by comparing the disparate conclusions of otherwise similar medical bodies that have assessed the vast and often contradictory body of evidence on penile circumcision.

Consider first the position of the American Academy of Pediatrics (AAP). In 2012, an eight-member AAP Task Force[28] asserted that the health benefits of NNMC outweigh the risk of surgical complications—albeit not to such an extent that the procedure could be recommended on medical grounds alone.[29] However, no recognized procedure for assigning weights, whether to benefits or risks, was used by the Task Force, so the basis for its conclusion was unclear. In a subsequent publication, Task Force members clarified that the benefits of NNMC were only "felt to outweigh the risks," and urged readers to "draw their own conclusions."[30] (Testifying at a government hearing on behalf of Germany's pediatric associations in late 2012, Dr. Wolfram Hartmann reported that the "AAP statement has been graded by almost all other pediatric societies and associations worldwide as being scientifically untenable.")[31]

In 2015, the Canadian Paediatric Society weighed in, concluding that the medical benefits and risks of NNMC were "closely balanced,"[32] while the Royal Dutch Medical Association

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[27] For a discussion of some of the controversy, see Jansen, M. (2016). Routine circumcision of infant boys: It's time to make progress through the common ground. *Journal of Paediatrics & Child Health,* 52(5), 477-479.

[28] It is concerning that there appears to have been financial, religious, and political conflicts of interest among the AAP Task Force members, though none were disclosed in the original statement. Task Force member Waldemar Carlo is a director of Mednax, the publicly-traded medical services company; although he failed to declare this financial conflict of interest in the original AAP policy statement, it was added to an AAP response to international critics: AAP Task Force on Circumcision (2013). Cultural bias and circumcision: the AAP Task Force on Circumcision responds. *Pediatrics*, 131(4), 801-804. Task Force member Andrew Freedman stated in an interview, "I circumcised [my son] myself on my parents' kitchen table on the eighth day of his life. But I did it for religious, not medical reasons. I did it because I had 3,000 years of ancestors looking over my shoulder." Merwin, T. (2012, September 19). Fleshing out change on circumcision. *Jewish Week*. Available at https://jewishweek.timesofisrael.com/fleshing-out-change-on-circumcision/ (accessed November 3, 2019). In a subsequent editorial, Freedman stated that "protecting" this parental option to circumcise "was not an idle concern" for the AAP Task Force members "at a time when there are serious efforts in both the United States and Europe to ban the procedure outright." Freedman, A. (2016). The circumcision debate: beyond benefits and risks. *Pediatrics*, 137 (5), e20160594. For extensive discussion of these matters, including analysis of the AAP response to its mostly European critics, see Earp & Shaw, *op cit.* note 11.

[29] AAP Task Force on Circumcision. (2012). Male circumcision. *Pediatrics,* 130(3), e756–785. For extensive critiques of the AAP analysis and further context, see note 11. See also Earp, B. D. (2015). Do the benefits of male circumcision outweigh the risks? A critique of the proposed CDC guidelines. *Frontiers in Pediatrics,* 3(18), 1–6.

[30] AAP Task Force on Circumcision. (2013). The AAP Task Force on Neonatal Circumcision: a call for respectful dialogue. *Journal of Medical Ethics*, 39(7), 442-443, 442.

[31] Hartmann, W. (2012). Expert Statement: Dr med. Wolfram Hartmann, President of Berufsverband der Kinder- und Jugendärzte, "for the hearing on the 26th of November 2012 concerning the drafting of a federal government bill." Cologne, Germany: Berufsverband der Kinder- und Jugendärzte (BVKJ. e.V.). Available at http://www.rz-etelsen.net/Hartmann%20-%20German%20Statement.pdf (accessed November 3, 2019).

[32] Canadian Paediatric Society. (2015). Newborn male circumcision. *Paediatrics & Child Health,* 20(6), 311–320. For further discussion, see Earp, B.D. (2015). Letter to the editor: strengths and weaknesses in the 2015 Canadian Paediatric Society statement regarding newborn male circumcision. *Paediatrics & Child Health,* 20(8), 433; Morris, B. J., Klausner, J. D., Krieger, J. N., Willcox, B. J., Crouse, P.D., & Pollock, N. (2016). Canadian

maintained that there "is no convincing evidence that [NNMC] is useful or necessary in terms of prevention or hygiene."[33] Meanwhile, the Royal Australasian College of Physicians, upon revisiting its 2010 policy in light of the AAP statement, affirmed that "the frequency of diseases modifiable by circumcision, the level of protection offered by circumcision and the complication rates of circumcision do not warrant routine infant circumcision in Australia and New Zealand."[34] Finally, the Danish Medical Association declared in 2016 that NNMC is "ethically unacceptable" on account of the known surgical risks, the lack of sufficient evidence of a "clear health benefit," and the permanency of the anatomical change.[35]

What these various positions suggest is that insofar as NNMC does confer some kind or degree of health-related benefit, it is far from clear that this benefit offsets even the strictly health-related risks, let alone in a decisive manner. Thus, even the AAP, whose policy among comparable organizations is the most favorable toward circumcision, does not actually recommend NNMC on grounds of health. But even if there were universal agreement among qualified medical authorities that the net health benefits of circumcision were as impressive as its strongest advocates claim (see Box 2),[36] this would not entail that NNMC—specifically—was medically, much less all-things-considered morally, justified.

This is because the strongest evidence of health benefit is derived from studies of adult, not newborn, circumcision; and this evidence primarily concerns risks pertaining to sexually

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Pediatrics Society position statement on newborn circumcision: a risk-benefit analysis revisited. *The Canadian Journal of Urology,* 23(5), 8495–8502; Robinson, J. L., Ann Jefferies, A., & Lacaze, T. (2017). Letter to the editor – Re: Canadian Pediatrics Society position statement on newborn circumcision: a risk-benefit analysis revisited. *The Canadian Journal of Urology,* 24(1), 8684–8687.

[33] The Royal Dutch Medical Association. (2010). Nontherapeutic circumcision of male minors. *KNMG*, 1–17. Available at https://www.knmg.nl/circumcision/ (accessed November 3, 2019).

[34] Royal Australasian College of Physicians, Paediatrics & Child Health Division. (2010). Circumcision of infant males. RACP. Available at https://www.racp.edu.au/docs/default-source/advocacy-library/circumcision-of-infant-males.pdf (accessed November 3, 2019); see also Pringle, K. (2014, April 4). Circumcision health risks and benefits - experts respond. *Science Media Center*. Available at http://www.sciencemediacentre.co.nz/2014/04/04/circumcision-health-risks-and-benefits-experts-respond/ (accessed November 3, 2019).

[35] Danish Medical Association. (2016). Lægeforeningens politik vedrørende omskæring af drengebørn uden medicinsk indication. Available at https://www.laeger.dk/laegeforeningens-politik-vedroerende-omskaering-af-drengeboern-uden-medicinsk-indikation (accessed November 3, 2019).

[36] See for instance Morris, B. J. (2007). Why circumcision is a biomedical imperative for the 21st century. *Bioessays,* 29(11), 1147-1158. Morris's typical claim, which has not been substantiated by anyone outside his research group, is that the benefits of circumcision outweigh the risks by a ratio of 100:1. But as Svoboda, Adler, and Van Howe recently noted, "no recognized procedure for objectively assigning weights to individual benefit or risks is used by Morris in his calculations [so] the ratio should not be taken seriously. For example, with no sound justification for how the higher number was obtained, in 2017 Morris et al. increased the ratio to 200 to 1, suggesting that the ratio is not scientifically meaningful." Svoboda, J. S., Adler, P. W., & Van Howe, R. S. (2019). Is circumcision unethical and unlawful? A response to Morris et al. *The Journal of Medical Law and Ethics* 7(1), 72–92; 84. The 200 to 1 assertion is made in Morris, B. J., Kennedy, S.E., Wodak, A.D., et al. (2017). Early infant male circumcision: systematic review, risk-benefit analysis, and progress in policy. *World Journal of Clinical Pediatrics*, *6(1)*, 89-102.

active persons, which does not include most children. Yet by the time a person does become sexually active, they will most likely be capable of consenting to circumcision if that is what they choose. They will also be capable of *withholding* their consent if they determine that circumcision is not in their medical (or other) best interests, and they can make this judgment on the basis of their actual—rather than hypothetical/future—health-related context and behavior.

In short, when a potential health benefit can be achieved either via (1) a non-voluntary surgical procedure performed on an individual to whom the benefit does not currently apply, and likely will not apply for many years (if at all); or (2) a similar surgical procedure undertaken voluntarily by the same individual when that individual is a position to decide whether the benefit is actually applicable (and if it is, whether it is worth the medical and non-medical risks given their known values/preferences and the available non-surgical alternatives); it is not normally justified, *even on medical grounds*, to proceed with option (1).[37]

As we noted in the Introduction, one potential health benefit that has received much attention in the recent literature is a partial reduction in the risk of female-to-male, heterosexually transmitted HIV as a result of voluntary medical male circumcision (VMMC). The main support for this claimed benefit stems from studies conducted in sub-Saharan Africa and published in the latter half of the 2000s.[38] At present, it is scientifically irresponsible to extrapolate from the findings of these studies on adult circumcisions performed in one epidemiological environment, to neonatal circumcisions performed in a dissimilar epidemiological environment.[39] Nevertheless, both "strong" and "weak" advocates of NNMC (Box 2) have appealed to the African RCTs in recent years to promote (in the case of strong advocates) or defend (in the case of weak advocates) non-voluntary circumcision of newborn males: not only in sub-Saharan Africa, where heterosexually transmitted HIV remains at epidemic levels, but also in the United States and other developed countries, where heterosexually transmitted HIV is relatively rare.[40] In the African context, a plausible motivation for the shift to infancy is the failure of VMMC programs to attract the number of men set by target quotas, despite the enormous effort and

[37] A counterargument to this view is that the surgical procedure in (1) is less medically risky than the similar surgical procedure in (2). We discuss this claim in a later section.
[38] Auvert et al., *op cit*. note 1; Bailey et al., *op cit*. note 1; Gray et al., *op cit*. note 1.
[39] For a discussion of why, see Bossio, J. A., Pukall, C. F., & Steele, S. S. (2014). A review of the current state of the male circumcision literature. *The Journal of Sexual Medicine,* 11(12), 2847–2864.
[40] Morris, B. J., & Krieger, J. N. (2015). The literature supports policies promoting neonatal male circumcision in North America. *The Journal of Sexual Medicine,* 12(5), 1305. But see Bossio, J. A., Caroline F. Pukall, C. F., & Steele, S. S. (2015). Response to: the literature supports policies promoting neonatal male circumcision in N. America. *The Journal of Sexual Medicine* 12(5), 1306–1307.

expense directed at supply-side "demand creation."[41] This shift is concerning from an ethical perspective, as infants, in contrast to unwilling adolescents or adults, are unable to refuse to the procedure.

Most other claimed health benefits similarly accrue after sexual debut. As a group of mostly European pediatricians and other medical authorities argued in response to the 2012 AAP policy statement on NNMC, there is only one potentially substantial health benefit that has "theoretical relevance in relation to infant male circumcision; namely, the possible protection against urinary tract infections in infant boys."[42] This potential benefit has never been studied via RCT, but observational and cohort studies cited by the AAP suggest that UTIs are anywhere between three and ten times more likely to occur in genitally intact male infants under 2 years of age than in circumcised ones.[43] A problem with interpreting such figures is that the rate of false positives may be higher for diagnoses among the former group than the latter.[44] But even if one takes such figures at face value, the difference between absolute and relative risk needs to be highlighted. According to the AAP's own estimates, UTIs affect roughly 1% of boys in the first two years of life regardless of circumcision status, suggesting that "the number needed to circumcise to prevent [one] UTI is approximately 100."[45] However, rather than permanently excising a sensitive genital structure from 100—or even far fewer—healthy boys to prevent one of them from acquiring a UTI, the same hypothetical UTI could be safely, successfully, and non-

[41] See Van Howe, R.S. (2018). Expertise or ideology? A response to Morris et al. 2016, "Circumcision is a primary preventive against HIV infection: critique of a contrary meta-regression analysis by Van Howe." *Global Public Health*, 13(12),1900-1918; **S**idler et al., *op cit*. note 5.

[42] Frisch et al., *op cit.* note 11, p. 796.

[43] AAP Task Force on Circumcision, *op cit.* note 29, p. e756.

[44] As the authors of the Canadian Paediatric Society statement on NNMC note, "urines obtained via a midstream or catheter specimen from an uncircumcised male are commonly contaminated by organisms under the foreskin. Evidence for this is that 9% of uncircumcised and 0.5% of circumcised asymptomatic males had bacteruria later verified by suprapubic urine collection to be falsely positive." Robinson et al., *op cit*. note 32, p. 8684, referring to Simforoosh, N., Tabibi, A., Khalili, S. A. et al (2012). Neonatal circumcision reduces the incidence of asymptomatic urinary tract infection: a large prospective study with long-term follow up using Plastibell. *Journal of Pediatric Urology,* 8(3), 320-323. See also Van Howe, R. S. (2005). Effect of confounding in the association between circumcision status and urinary tract infection. *Journal of Infection*, 51(1), 59-68.

[45] AAP Task Force on Circumcision, *op cit.* note 29, p. e767. Other credible estimates of the number-needed-to-prevent range from 25 to 100 – see Eisenberg, M. L., Galusha, D., Kennedy, W. A., & Cullen, M. R. (2018). The relationship between neonatal circumcision, urinary tract infection, and health. *The World Journal of Men's Health*, 36(3), 176-182 – to as many as 195; see To, T., Agha, M., Dick P. T., Feldman W. (1998). Cohort study on circumcision of newborn boys and subsequent risk of urinary-tract infection. *Lancet*, 352, 1813–1816. Some "lifetime" estimates have been published including UTIs potentially acquired between sexual debut and death, but these fall outside the scope of our analysis.

invasively treated with antibiotics in most cases.[46] This is the standard of care for girls, who are about 4 to 8 times more likely than boys to acquire a UTI by age 5 years, according to the AAP.[47]

Balanitis (or inflammation of the glans penis) is also thought to occur less often in circumcised males, although it may actually be more common in such males while they are still in diapers.[48] According to the UK National Health Service, balanitis is "not usually serious" and "most cases [are] easily treated with good hygiene and creams [or] ointments."[49] Again, it is not clear why indiscriminately removing genital tissue from healthy infants to reduce—potentially—the risk of a typically non-serious and easily treatable source of temporary discomfort should even be entertained. Phimosis, the inability to fully retract the prepuce over the glans of the penis, necessarily occurs less frequently in circumcised males who lack a prepuce.[50] However, in young boys, a non-retractile foreskin is normal as the foreskin is fused to the glans at birth, after which the two structures gradually separate during early childhood. In approximately 50% of cases, this process will be complete by ten years of age, but it can continue until after the onset of puberty.[51] For this reason, phimosis, as such, is difficult to diagnose accurately before adolescence. Moreover, if a lack of ability to retract the foreskin does create problems for a child, this can usually be treated in a less invasive way than by excising the foreskin altogether: topical creams and gentle stretching will resolve the majority of problematic cases with no cutting of tissue

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[46] Fisher, D. J. (2019). Pediatric urinary tract infection treatment & management. *Medscape*. Available at https://emedicine.medscape.com/article/969643-treatment (accessed November 3, 2019): "Most cases of uncomplicated UTI respond readily to outpatient antibiotic treatments without further sequelae."

[47] What is more, "most children who have one UTI will not have another." American Academy of Pediatrics. (2018). Recurrent urinary tract infections (UTIs) in children. From the AAP Section on Urology. Available at https://www.healthychildren.org/English/health-issues/conditions/genitourinary-tract/Pages/Urinary-Tract-Infections-in-Teens.aspx (accessed November 3, 2019).

[48] Van Howe, R. S. (2007). Neonatal circumcision and penile inflammation in young boys. *Clinical Pediatrics,* 46, 329-333. See also Escala, J. M., & Rickwood, A. M. K. (1989). Balanitis. *British Journal of Urology*, 63(2), 196-197. According to these authors, balanitis "is usually associated with a prepuce which is partly or completely non-retractable. It does not cause phimosis and no single pathogen is involved. Most boys suffer a single episode and circumcision is indicated only for those with recurrent, troublesome attacks."

[49] NHS. (2017). Balanitis. UK National Health Service. Available at https://www.nhs.uk/conditions/balanitis/ (accessed November 3, 2019).

[50] The incidence of pathologic phimosis in intact males (0.5% to 1%) appears to be similar to the incidence of preputial stenosis/phimosis following circumcision (0.3-1.7%). See Shankar, K. R., & Rickwood, A. M. K. (1999). The incidence of phimosis in boys. *BJU International*, 84, 101-102; Kaweblum, Y. A., Press, S., Kogan, L., Levine, M., & Kaweblum, M. (1984). Circumcision using the Mogen clamp. *Clinical Pediatrics*, *23*, 679-682; Stenram, A., Malmfors, G., & Okmian, L. (1986). Circumcision for phimosis — indications and results. *Acta Paediatrica Scandinavia*, 75, 321-323; Van Howe, R. S. (1997). Variability in penile appearance and penile findings: a prospective study. *British Journal of Urology*, 80, 776-782.

[51] Kayaba, H., Tamura, H., Kitajima, S., Fujiwara, Y., Kato, T., & Kato, T. (1996). Analysis of shape and retractibility of the prepuce in 603 Japanese boys. *Journal of Urology*, 156, 1813-1815; Øster, J. (1968). Further fate of the foreskin. Incidence of preputial adhesions, phimosis, and smegma among Danish schoolboys. *Archives of Disease in Childhood*, 43, 200-203.

required, and most of the remainder can be addressed with foreskin-preserving preputioplasty (for instance, a limited dorsal slit with transverse closure).[52]

Recurrent pathological phimosis is thought to be a risk factor for penile cancer, which may explain why (especially early) circumcision has been associated with a lower relative risk of contracting this rare cancer.[53] According to recent estimates, "penile cancer is one of the rarest malignancies for which site-specific data are available in cancer registries. In 2003–2007, penile cancers accounted for 0.15% of all incident cancers in the United States, thus ranking 43rd in incidence among all site-specific cancers in US males."[54] In fact, penile cancer "is not among the 10 most common malignancies in males in any national cancer registry; even in high-incidence regions of South America, Africa, and Asia, the risk of developing penile cancer before age 75 years is below 0.4%."[55] In line with this, the AAP estimates that between 909 and 322,000 circumcisions would be required to prevent a single case.[56]

A similar analysis applies to most of the health benefits commonly cited by NNMC advocates. We do not have room to cover them all here. Instead, we refer the reader to more exhaustive critical surveys of the claimed health benefits of NNMC that have recently been performed by others.[57] Our point in this section has not been to imply that NNMC has no statistical health-related benefits. Rather, it has been to emphasize that (1) the likelihood and

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[52] In a large prospective study, a total of 1185 boys with a diagnosis of pathological phimosis were treated with fluticasone proprionate. Successful results with no side effects were achieved in 1079 (91.1%) patients including those diagnosed with mild balanitis xerotica obliterans. Zavras, N., Christianakis, E., Mpourikas, D., & Ereikat, K. (2009). Conservative treatment of phimosis with fluticasone proprionate 0.05%: a clinical study in 1185 boys. *Journal of Pediatric Urology,* 5(3), 181-185. Regarding preputioplasty, see, e.g., Hotonu, S., Mohamed, A., Rajimwale, A., & Gopal, M. (2019). Save the foreskin: outcomes of preputioplasty in the treatment of childhood phimosis. *The Surgeon,* in press.

[53] See, e.g., Larke, N. L., Thomas, S. L., dos Santos Silva, I., & Weiss, H. A. (2011). Male circumcision and penile cancer: a systematic review and meta-analysis. *Cancer Causes & Control,* 22(8), 1097-1110. These authors note, however, that in 2 of the 8 studies included in their analysis, "the protective effect of childhood/adolescent circumcision on invasive cancer no longer persisted when analyses were restricted to boys with no history of phimosis."

[54] Frisch, M. (2017). Penile cancer. In M. Thun et al. (Eds.), *Cancer epidemiology and prevention* (ch. 55), Oxford: Oxford University Press.

[55] *Ibid*, note 54.

[56] AAP Task Force on Circumcision, *op cit.* note 29, p. e768.

[57] See for example Dave, S., Afshar, K., Braga, L. H., & Anderson, P. (2018). Canadian Urological Association guideline on the care of the normal foreskin and neonatal circumcision in Canadian infants (full version). *Canadian Urological Association Journal*, 12(2), E76-E99. The authors fail to find that the health benefits of NNMC outweigh the medical risks, concluding that from an "overall societal perspective, given our healthcare system and the socioeconomic and educational status of our population, universal neonatal circumcision is not justified based on the evidence available." The most recent evidence-based review from any non-partisan group or organization of which are aware is the one from Evidence Based Birth: Dekker, R., & Bertone, A. (2019). Evidence and ethics on: circumcision. *Evidence Based Birth*. Available at https://evidencebasedbirth.com/evidence-and-ethics-on-circumcision/ (accessed November 3, 2019). These authors state: "After an extensive review of the literature, we've concluded that there is no compelling evidence to justify routine male infant circumcision on medical grounds."

magnitude of these benefits is scientifically controversial; (2) most medical bodies that have studied NNMC have failed to conclude that the health-related benefits outweigh even the strictly health-related risks or harms; (3) most of the benefits cited by NNMC advocates do not apply until sexual debut, which is typically at, near, or after the time at which the vast majority of individuals could consent to circumcision if that is what they wanted; (4) the main health benefits said to apply before sexual debut have a low absolute risk of occurring, are typically non-serious, and can usually be effectively treated without tissue loss; and (5) nearly all of the health benefits attributed to circumcision can be achieved in less invasive ways that do not involve non-voluntary genital cutting.

B. Costs

We turn next to the question of costs or drawbacks to NNMC, which we have divided into three categories: ***general harms***, ***medical complications***, and ***financial costs***. These costs are discussed separately because in Anglophone Western countries, even those with low or declining rates of NNMC, it is commonplace to discuss circumcision solely in terms of its medical benefits and surgical risks. The *prima facie* value of the prepuce itself (see Box 3), the state of being genitally intact, and/or the ability to decide for oneself about whether to undergo an elective genital procedure—all of which are necessarily lost to NNMC—are not typically given as much attention. But insofar as an individual does place a positive value on any of those factors (the prepuce, genital intactness, or having a choice), NNMC is a harm to them even if there are *no* surgical complications.[58] The magnitude of the harm, in turn, depends on the degree of value placed on the factors, which cannot be known in infancy or early childhood and is likely to be highly individually variant.

Another problem with current conceptions of harm relating to NNMC is that anatomically normal and benign attributes of the foreskin are sometimes implied to have negative value. For example, according to the 2012 AAP statement, "Penile wetness (defined as the observation of a diffuse homogeneous film of moisture on the surface of the glans and coronal sulcus) is considered a marker for poor penile hygiene and is more prevalent in uncircumcised than in circumcised men."[59] In support of this view, the AAP Task Force cites a study finding a higher

[58] Svoboda, J. S. (2017). Nontherapeutic circumcision of minors as an ethically problematic form of iatrogenic injury. A*MA Journal of Ethics,* 19(8), 815-824.
[59] AAP Task Force on Circumcision, *op cit*. note 29, p. e763.

rate of "penile wetness" among intact than circumcised men.[60] The study does not explain how penile wetness constitutes "poor penile hygiene," nor does it define what is meant by the latter concept. It merely notes, as does the AAP Task Force, that other authors have used "penile wetness" as a proxy for "poor penile hygiene." Neither the authors of the study nor the AAP Task Force seem aware that penile wetness in genitally intact males is normal, just as labial wetness in genitally intact females is normal. The glans penis, like the glans clitoris, evolved though natural selection to be an internal organ that is protected and lubricated by a prepuce, akin to the relationship of the eye to the eyelid.[61]

Box 3. A brief overview of the human prepuce

The prepuce is a "common anatomical structure of the male and female external genitalia of all human and non-human primates." In humans, the penile and clitoral prepuces are identical in early fetal development and remain indistinguishable in some intersex individuals. The prepuce is an "integral, normal part of the external genitalia that forms the anatomical covering of the glans penis and clitoris," thereby internalizing each and "decreasing external irritation and contamination." In the case of the penile prepuce, an additional function is to protect the urinary opening from abrasion, as this runs through the penile, but not the clitoral glans. In both cases, the prepuce is "a specialized, junctional mucocutaneous tissue which marks the boundary between mucosa and skin … similar to the eyelids, labia minora, anus and lips." The "unique innervation of the prepuce establishes its function as an erogenous tissue."[62]

General harms

Circumcision removes one-third to one-half of the motile skin system of the penis.[63] This is tissue that many individuals consider to be erogenous, as it provides a range of specific sensations that can contribute to the subjective experience of sexual enjoyment.[64] The "gliding" motion of the

[60] O'Farrell, N., Morison, L., & Chung, C. K. (2007). Low prevalence of penile wetness among male sexually transmitted infection clinic attendees in London. *Sexually Transmitted Diseases,* 34(6), 408–409.
[61] Cold & Taylor, *op cit*. note 14.
[62] Quotations from Cold & Taylor, *op cit*. note 14.
[63] Taylor, J.R., Lockwood, A.P., & Taylor, A.J. (1996). The prepuce: specialized mucosa of the penis and its loss to circumcision. *British Journal of Urology,* 77, 291-295.
[64] *Ibid.* See also Martín-Alguacil, N., Cooper, R. S., Aardsma, N., Mayoglou, L., Pfaff, D., & Schober, J. (2015). Terminal innervation of the male genitalia, cutaneous sensory receptors of the male foreskin. *Clinical Anatomy*, 28(3), 385-391; Ball, P. J. (2006) A survey of subjective foreskin sensation in 600 intact men. In *Bodily integrity and the politics of circumcision* (pp. 177–188). New York: Springer; Wong, D. P., Morrison, B. F., Mayhew, R. G., Reid, G. A., & Aiken, W. D. (2015). A delayed foreskin-sparing approach to the management of penile fractures in uncircumcised Jamaican men. *International Journal of Surgery Case Reports*, 17, 65-68; Kim, D., & Pang, M. G. (2007). The effect of male circumcision on sexuality. *BJU International,* 99(3), 619-622.

foreskin back and forth over the glans during sexual intercourse or masturbation is permanently disabled by NNMC, changing the very mechanics of penile stimulation;[65] and part or all of the frenulum is often removed.[66] Studies employing objective measures consistently show that the foreskin is the most sensitive part of the organ to light touch.[67] Moreover, scar tissue may form around the circumference of the penis following circumcision, which apart from being unsightly in some cases, may also be less capable of supplying varied sensation.[68] Finally, pain is an unavoidable consequence of circumcision, not only during the procedure itself but also throughout the healing process. Because general anesthesia is considered too risky for neonates, the most effective form of pain control is not available to them (whereas it is available for adults). Instead, local anesthetics must be used. However, the best of these approved for use in neonates still requires making injections into the base of the penis prior to circumcision, which can itself be painful.[69]As Bellieni et al. noted after a review of the literature, "there is no such thing as a pain-free circumcision."[70] A further consideration is that adults can communicate their pain levels

[65] For a demonstration, see MFN24. (2015). Gliding motion of the foreskin. *Wikimedia Commons*. Available at https://commons.wikimedia.org/wiki/File:Foreskin_in_motion.gif (accessed November 3, 2019).
[66] See Shenoy, S. P., Marla, P. K., Sharma, P., Bhat, N., & Rao, A. R. (2015). Frenulum sparing circumcision: step-by-step approach of a novel technique. *Journal of Clinical and Diagnostic Research*, 9(12), PC01–PC03.
[67] Sorrells, M. L., Snyder, J. L., Reiss, M.D., Eden, C., Milos, M. F., Wilcox, N., & Van Howe, R.S. (2007). Fine-touch pressure thresholds in the adult penis. *BJU International*, 99(4), 864-869; Bossio J. A., Pukall, C. F., & Steele, S. S. (2016). Examining penile sensitivity in neonatally circumcised and intact men using quantitative sensory testing. *The Journal of Urology,* 195(6), 1848–1853. For critical discussion and contextualization of these findings, see Earp, B. D. (2016). Infant circumcision and adult penile sensitivity: implications for sexual experience. *Trends in Urology & Men's Health,* 7(4), 17–21; Van Howe, R. S., Sorrells, M. L., Snyder, J. L., Reiss, M. D., & Milos, M. F. (2016). Re: Examining penile sensitivity in neonatally circumcised and intact men using quantitative sensory testing. *The Journal of Urology,* 196(6), 1824; Rotta, A. T. (2016). Re: "Examining penile sensitivity in neonatally circumcised and intact men using quantitative sensory testing." *The Journal of Urology*, 196(6), 1822-1823.
[68] Cold & Taylor, *op cit*. note 14. See also Fahmy, M. A. B. (2019). Nonaesthetic circumcision scarring. In Fahmy, M. A. B. (ed)., *Complications in male circumcision* (pp. 99-134). Amsterdam: Elsevier.
[69] Banieghbal, B. (2009). Optimal time for neonatal circumcision: an observation-based study. *Journal of Pediatric Urology*, 5(5), 359-362. The ring block injections administered prior to circumcision in this study resulted in observable pain responses above a pre-defined threshold in 31% of <1 week-old infants and 85% of 1–5 week-old infants. That is, simply being administered the anesthesia was painful for many infants at a clinically significant level. Moreover, among 1–5 week-old infants, 71% experienced above-threshold pain during the actual circumcision. Some research suggests that infants experience pain, if anything, more acutely than adults: "The newborn is inexperienced in filtering noxious stimuli, so the full impact of pain, without adaptive measures, is delivered. In older children and adults, pain is attenuated by filters developed with experience by the nervous system. This difference in our perception may be explained by older [individuals] who can articulate their level of pain, whereas infants are reduced to screams that the circumcising physician may ignore" Van Howe, R. S., & Svoboda, J. S. (2008). Neonatal pain relief and the Helsinki Declaration. *The Journal of Law, Medicine & Ethics*, 36(4), 803-823. See also Fitzgerald, M. (2015). What do we really know about newborn infant pain? *Experimental Physiology*, 100(12), 1451-1457.
[70] Bellieni, C. V., Alagna, M. G., & Buonocore, G. (2013). Analgesia for infants' circumcision. *Italian Journal of Pediatrics*, 39(38),1-7. It is notable that a separate AAP Task Force, charged with assessing the literature on infant pain rather than circumcision, concluded that: "exposure to repeated painful stimuli early in life is known to have short- and long-term adverse sequelae [including] physiologic instability, altered brain development, and abnormal

clearly and manage their own pain control needs as the circumcision wound heals. Pre-verbal infants, by contrast, whose wound—unlike that of an adult—will be exposed to urine and feces in a diaper, cannot as effectively communicate their discomfort and must rely on others to change their bandaging, administer analgesic drugs, and so on, in a timely and appropriate manner.

Most of the above outcomes, which may reasonably be judged to constitute harms,[71] could apply to circumcisions performed at any stage of life. However, as with the pain issue, some potential harms of circumcision may be greater in infancy or early childhood compared to after an age of consent. First, on a psychological level, there is the sheer removal of choice concerning how an intimate part of one's body should look or function. Some individuals regard this aspect of NNMC as harmful in and of itself.[72] Second, on a more physical level, because the infant penis is so small, the physician performing the circumcision may remove too much tissue, possibly contributing to tight or painful erections in adulthood, without being able to anticipate or fully account for this adverse outcome.[73] It may also be more likely that the frenulum would be cut away with those devices most commonly used to perform a neonatal or infant circumcisions (e.g.,

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neurodevelopment, somatosensory, and stress response systems, which can persist into childhood," and recommended that "every health care facility caring for neonates should implement [a] pain-prevention program that includes strategies for minimizing the number of painful procedures performed." AAP Committee on Fetus and Newborn and Section on Anesthesiology and Pain Medicine (2016). Prevention and management of procedural pain in the neonate: an update. *Pediatrics*, 137(2): e20154271.

[71] Of course, one and the same outcome may be judged to be—or experienced as—a harm for one individual, while, for another individual with different preferences, it could be judged to be a benefit. Once again, such disagreement weighs in favor of allowing the individual to decide about their own genitals in terms of whether they should be permanently modified. For extensive discussions, see Earp, B. D, & Darby, R. (2017). Circumcision, sexual experience, and harm. *University of Pennsylvania Journal of International Law,* 37(2-online), 1–57; Earp, B. D. (2017). Gender, genital alteration, and beliefs about bodily harm. *Journal of Sexual Medicine.* 14(5), Supp. 4, e225 (associated video presentation titled "Circumcision – a sexual harm?" is available at https://www.youtube.com/watch?v=SB-2aQoTQeA).

[72] Hammond, T. & Carmack, A. (2017). Long-term adverse outcomes from neonatal circumcision reported in a survey of 1,008 men: an overview of health and human rights implications. *International Journal of Human Rights*, 21(02), 189-218. Bossio J.A., & Pukall C.F. (2018). Attitude toward one's circumcision status is more important than actual circumcision status for men's body image and sexual functioning. *Archives of Sexual Behavior,* 47(3), 771-781.

[73] Hammond & Carmack. *op cit.* note 72, pp. 199-200. See also Kaplan, G. W. (1983). Complications of circumcision. *Urologic Clinics of North America*, 10(3), 543-549, referring to skin bridges as further potential complication of NNMC that may cause tight or painful erections. The size of the infant penis also may also explain why "partial glans amputation, buried penis and total penectomies" are complications that are primarily limited to early circumcisions. Ungar-Sargon, E. (2015). On the impermissibility of infant male circumcision: a response to Mazor (2013). *Journal of Medical Ethics*, *41*, 186-190. For supportive data, see PEPFAR (2019). PEPFAR 2020 country operational plan guidance for all PEPFAR countries. PEPFAR. Available at https://www.state.gov/wp-content/uploads/2019/11/2019-11-25-COP20-Guidance-Full-Consolidated_Public-2-1.pdf (accessed November 3, 2019). Likewise, deaths following circumcision performed on infants are reported on occasion, including in developed countries, whereas deaths following the procedure performed under similar conditions on adults are virtually unheard of. See Earp, B. D., Allareddy, V., Allareddy, V., & Rotta, A.T. (2018). Factors associated with early deaths following neonatal male circumcision in the United States, 2001- 2010. *Clinical Pediatrics,* 57(13), 1532–1540; Paediatric Death Review Committee: Office of the Chief Coroner of Ontario. (2007). Circumcision: a minor procedure? *Paediatrics & Child Health*, 12, 311-312.

the Gomco clamp, Mogen clamp, or Plastibell), which less discriminately remove tissue around the diminutive infant glans. In voluntary circumcisions, by contrast—that is, when the penis has reached its full size—it is easier for the physician to cut around the frenulum, leaving it mostly or entirely intact.[74]

In terms of data collection and estimation of complications, problems associated with NNMC may be less likely to be recognized or reported: some individuals may go their entire lives thinking that a negative outcome is normal, or fail to associate the problem with their circumcision, since this occurred prior to their ability to make a conscious before-and-after comparison. Finally, the infant foreskin is normally fused to the head of the penis by a membrane, whereas the foreskin of a fully developed penis is more likely to be detached. The authors of a study comparing neonatal circumcision with that of older boys—to be discussed further below—explain the surgical significance of this distinction:

> In the neonatal period, irrespective of the applied method of circumcision, a forced and traumatic degloving of the foreskin from the meatus, glans, and sulcus cannot be avoided … Initially, the inner surface of the prepuce and the epithelium of the glans are fused, and separation of these two surfaces develops progressively [which] explains the need for forceful and traumatic degloving in neonatal circumcision. Degloving leads to a perimeatal inflammation, which can result in meatal deformity with eventual stenosis. Development of scar tissue with meatal deformity and stenosis [in our study] was secondary to the traumatic excoriation and was not merely due to exposure and irritation by the diaper of the fragile and under-keratinized glans epithelium.[75]

Even from a purely technical perspective, then, there are at least some costs to performing circumcision on a small and underdeveloped penis that do not apply to circumcision performed on a penis that has reached its full size and final configuration. Moreover, any long-term complications that are incurred in infancy or early childhood will affect the individual for a longer period of time and hence a greater proportion of their life compared to the same complication incurred in adolescence or adulthood.

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[74] On the importance of allowing the penis to reach its full size more generally, consider the analogous case of elective labiaplasty in females. According to the Royal College of Obstetricians and Gynaecologists (RCOG), "Even though children aged 16 or over can consent to surgical procedures," medically unnecessary female genital cutting procedures "should not normally be carried out on women and girls under 18 years of age, irrespective of consent, because full genital development is not normally achieved before the age of 18." RCOG Ethics Committee. (2013). Ethical opinion paper: Ethical considerations in relation to female genital cosmetic surgery (FGCS). RCOG. Available at https://www.rcog.org.uk/globalassets/documents/guidelines/ethics-issues-and-resources/rcog-fgcs-ethical-opinion-paper.pdf (accessed November 3, 2019).

[75] Machmouchi, M., & Alkhotani, A. (2007). Is neonatal circumcision judicious? *European Journal of Pediatric Surgery*, 17(04), 266-269, 268.

Medical Complications

The strictly medical complications of circumcision are often said to be fewer and less serious in infancy, especially the neonatal period, than at any other stage of life. As "strong" NNMC advocates (see Box 2) Brian Morris *et al.* state in a review, "Evidence clearly shows that circumcision in infancy carries fewer risks of complications than circumcisions performed in childhood or later in life."[76] Somewhat incongruously, however, they go on to argue (emphasis added):

> A recent systematic review found **a median complication frequency of 1.5%** among studies of neonatal or infant circumcision, compared to 6% among studies of children aged one year or older … In the large RCTs of adult MC, **complications were seen in 1.7-3.8%**; these were virtually all mild or moderate and were effectively treated.[77]

In other words, according to these authors, while complications from circumcision in early childhood (ages 1 year and older) occur about 4 times as frequently as in the neonatal or infant period—assuming that the cited data are correct; more on this below—there does not appear to be a very large absolute difference between complications arising from such neonatal or infant circumcision and *adult* circumcision, which is the more ethically relevant comparison in terms of the capacity to consent. In fact, even the relative difference in complication rates appears to be modest: the complication rate for adult circumcision, according to the authors, is between 1.13 and 2.53 times that of NNMC, with "virtually all" of the complications associated with the adult form being "mild or moderate" and "effectively treated." If that is the case, then the marginally greater alleged complication rate in adults (in absolute terms, between 0.2% and 2.3%) is arguably of little medical or ethical significance; whereas the capacity of the adult, compared to the infant, to consent to the procedure is of enormous ethical significance.

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[76] Morris et al., *op cit*. note 13.

[77] *Ibid.* Apart from this article by Morris et al., one of the most common sources of the claim that circumcision is more risky after infancy is the study by El Bcheraoui, C., Zhang, X., Cooper, C. S., Rose, C. E., Kilmarx, P. H., & Chen, R. T. (2014). Rates of adverse events associated with male circumcision in US medical settings, 2001 to 2010. JAMA Pediatrics, 168(7), 625-634. These authors claim that compared with boys circumcised at <1 year, the incidences of probable adverse events were "approximately 20-fold and 10-fold greater for males circumcised at age 1 to 9 years and at 10 years or older, respectively." However, these estimates were indirectly inferred from a retrospective comparison of hospital billing claims that did not distinguish between voluntary/non-voluntary or therapeutic/non-therapeutic circumcisions nor adjust for potential confounding with concurrent procedures. Moreover, the window of analysis employed was likely too short to capture one of the most common late-occurring complications associated with newborn circumcision, namely, meatal stenosis, such that "the observed low frequency of urethral strictures (0.01%) – and therefore the low overall frequency of complications for boys circumcised in infancy (0.4%) – most likely reflects a combination of insensitive research data and insufficient follow-up" (Frisch & Earp, op cit. note 21, p. 633).

Even so, the evidence cited by Morris *et al.* in support of their claims regarding an increased risk for adults does not come from well-controlled studies. Instead, it comes from studies of non-concurrent populations circumcised with different techniques and evaluated with disparate methodologies, using inconsistent definitions and thresholds for identifying complications. Very few studies have made a direct comparison between the complication rates of neonatal versus later circumcision within a single population, and those that have have yielded mixed results. One study from Iran, for example, found no statistically significant difference in late complications (defined as excessive residual foreskin, excessive foreskin removal, meatal stenosis, granuloma, penile rotation, and secondary chordee) when comparing circumcisions performed by medically untrained personnel in the neonatal period (5.68%) with those performed in later childhood (7.44%). However, more than 70% of the latter circumcisions took place between the ages of 2 and 12, with 12 being the maximum age of any child included in the study. Since this is below the age of consent in most jurisdictions, it is also below the age-range of the relevant comparison class for our purposes.[78]

Two other studies from Pakistan found a lower overall complication rate in neonates compared to older children, but similarly failed to make the ethically relevant comparison: the maximum age at which the circumcisions took place across both studies was 5 years old.[79] Nevertheless, in one of the studies, the authors reported a greater percentage of buried glans cases

[78] Yegane, R.-A., Kheirollahi, A.-R., Salehi, N.-A., Bashashati, M., Khoshdel, J.-A., & Ahmadi, M. (2006). Late complications of circumcision in Iran. *Pediatric Surgery International*, 22, 442-425. A similar limitation applies to a later Turkish study, which had a maximum age of 14 years: Bicer, S., Kuyrukluyildiz, U., Akyol, F., Sahin, M., Binici, O., & Onk, D. (2015). At what age range should children be circumcised? Iran Red Crescent Medical Journal, 17(3), e26258. These authors found no statistically significant difference in surgical complication rates between children circumcised at ages <1 year, 1-7 years, and 7-14 years. There were, however, fewer recorded complications due to anaesthesia in the <1 year group, but these children were circumcised with local anaethesia only, whereas general anaesthesia was used among the older children, with the majority complication being the temporary experience of double vision. We discuss the ethical significance of the use of local versus general anaesthesia elsewhere in the paper.

[79] Moosa, F. A., Khan, F. W., & Rao, M. H. (2010). Comparison of complications of circumcision by 'Plastibell device technique' in male neonates and infants. *Journal of the Pakistan Medical Association*, 60, 664-667. Razzaq, S. et al. (2018). Safety of the Plastibell circumcision in neonates, infants, and older children," *International Journal of Health Sciences* 12(5) 10–13. Similar findings, with similar limitations (e.g., maximum age of 8.5 months in the non-newborn comparison group) were reported in a retrospective study of 130 circumcisions performed by a single pediatric urologist based in the United States: Horowitz, M., & Gershbein, A. B. (2001). Gomco circumcision: When is it safe? Journal of Pediatric Surgery, 36(7), 1047-1049. More recently, a complication rate of 1.5% was reported for neonates over a five-year period compared to 2.9% in non-neonates; but again, the study was retrospective and poorly controlled, and the non-neonate group was not old enough to consent (max age < 5 years): Hung, Y. C., Chang, D. C., Westfal, M. L., Marks, I. H., Masiakos, P. T., & Kelleher, C. M. (2019). A longitudinal population analysis of cumulative risks of circumcision. Journal of Surgical Research, 233, 111-117. A prospective Indian study reported a 4.16% complication rate in neonates circumcised with the Plastibell device compared to 20% in infants, with the same wrong-comparison-group problem (max age 1 year): Shinde, N. D., Moinuddin, M., & Danish, A. O. (2018). Plastibell circumcision in neonates and infants at tertiary care centre. International Surgery Journal, 5(4), 1488-1491. And so on.

(3.21%) in neonates (mean age 15 days) than in infants (mean age 3 months) or older children (mean age 2.3 years) (1.24% and 1.92% respectively), "which may be due to the small size of [the neonatal] penis and adhesions."[80] Again, these findings suggest that different kinds of complications may occur at different rates depending on the age of the individual, with some being higher (or more serious) at a given age and others being lower (or less serious).

A final study, from Saudi Arabia, had a relatively well controlled prospective cohort design, and found a significantly higher rate of "serious complications" for neonatal circumcision compared to circumcision of older infants, regardless of the method used (bone cutter, Gomco clamp, Plastibell device). The authors concluded from this that "neonatal circumcision should not be recommended."[81] However this study, too, did not compare neonatal (or infant) circumcision with circumcision after an age of consent: the maximum age in the comparison group was 5 months old.

No studies to our knowledge have yet been performed comparing the circumcision of neonates or infants with that of consenting older adolescents or adults in the same population, properly controlling for the experience of the surgeon, the method used, the setting, and the reason for circumcision. This last factor is perhaps the most important. Since it is unusual, especially in the developed world, for adults to request a circumcision unless they already have a pre-existing medical condition relating to their penis, some of the greater apparent burden of complications in adults (on average) may be related to this factor. NNMC, by contrast, is usually performed on a normal, healthy penis. Some complications that do occur may be missed by the parents, and the infant will be in no position to recognize or to report these. The only other party who might report any complications arising from NNMC is the medical practitioner (who has an interest in not doing so).

That being said, recent findings from a relatively controlled comparison based on millions of cases, with all circumcisions performed as part of the ongoing VMMC program described above, suggest the very opposite of the conclusion drawn by Morris *et al*. in their review. According to the U.S President's Emergency Plan for AIDS Relief (PEPFAR), "Complications continue to be reported more commonly among those under age 15 at VMMC, *especially in infants*." In particular, fully 100% of glans injuries and 90% of fistulas reported through their Notable Adverse Events (NAE) reporting system occurred in children circumcised at ages 14 and younger. And the overall ratio NAEs to circumcisions performed was five times higher in infants

[80] *Ibid*, p. 12.
[81] Machmouch & Alkhotani, *op cit*. note 75, p. 266.

(15.3 per 100,000) than in young adolescents (2.9 per 100,000), which in turn was about twice the ratio for individuals circumcised at ages 15 years and above (1.6 per 100,000). As a result of these findings, PEPFAR announced in late 2019 that it will no longer fund infant circumcisions as a part of the VMMC program, citing the "NAE review, the severity of glans injuries and fistulas when they occur, and expected timing of pubertal development." The minimum eligibility criteria for VMMC have been updated to 15 years of age or Tanner Stage 3 sexual development.[82]

Even in developed countries, severe complications associated with NNMC may be more common than suggested by its proponents' conservative estimates. At a single children's hospital in Birmingham, England, for instance, cases of "life threatening hemorrhage, shock or sepsis" resulting from infant circumcision ranged from 0 to 11 per year between 2002 and 2011.[83] Since these figures were obtained due to a special freedom of information request and otherwise would not have been made public, it is unclear whether the complication rates in the published literature reflect the full scope of the problem. Finally, it has been suggested that "the severity of complications is far more important than the rate. Although necrotizing fasciitis, amputation of the penis and death are incredibly rare," they may be more common in infancy than in adulthood (assuming a similar clinical context and a healthy organ); and "some would consider it unethical [to] advocate for an elective procedure that may or may not benefit an individual [non-consenting] patient yet has *any* potential to result in such devastating outcomes."[84]

Financial costs

Morris *et al.* point to a lower financial cost of performing circumcision in infancy, estimating in 2012 a cost of between $165 and $257 in infancy, compared to $1,800-2,000 for circumcision in adolescence or adulthood. They acknowledge, however, that the cost for older males "can be reduced by insisting on a local anesthetic, since a general anesthesiologist's fees can be considerable."[85] As noted, general anesthesia is typically contra-indicated in neonates, which appears to be the main reason why neonatal circumcisions are generally less expensive overall.

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[82] PEPFAR, *op cit*. note 73, pp. 171-2, emphasis added. For further discussion, see Earp & Darby, *op cit*. note 71.
[83] Checketts, R. (2012). Response to freedom of information request, FOI/0742. Birmingham Children's Hospital, NHS Foundation Trust. Available at http://www.secularism.org.uk/uploads/foi-bch-response-received-260612.pdf (accessed November 3, 2019). For a recent discussion of some of the challenges obtaining FOI data on circumcision complications, see Fox, M., Thomson, M., & Warburton, J. (2018). Non-therapeutic male genital cutting and harm: law, policy and evidence from UK hospitals. *Bioethics*, 33(4), 467-474.
[84] Robinson et al., *op cit*. note 32, p. 8684, emphasis added. Against this view, it could be argued that certain risks associated with a lack of circumcision, such penile cancer, may also be devastating. However, these typically already-low risks can almost always be reduced or eliminated without genital surgery, which carries its own risks, whereas the most devastating injuries that may be caused by circumcision (e.g., partial or complete amputation of the glans) are typically more directly attributable to the surgery itself.
[85] Morris et al., *op cit.* note 13, p. 8.

Morris *et al.* state that if adolescent and adult males opted for circumcision under local anesthesia, the discrepancy in cost would not be as great. Consistent with this, a more recent, informal estimate suggests that when the procedure is not covered by health insurance, "circumcision for a newborn infant typically costs $150-$400 for the doctor fee, and possibly an additional facility fee, which can increase the total to $800 or more," whereas "circumcision for an older child or adult male typically costs $800-$3,000 or more." As a characteristic example, the fees from one circumcision clinic are listed as "$850 for children 1 to 17 years, $1,500 for adults if local anesthesia is used and $3,000 for adults if general anesthesia is used."[86] Assuming for the sake of argument that these figures are in the right ballpark, the difference in cost between a medically unnecessary newborn circumcision ($800+ on the high end, including a facilities fee) and that of an older adolescent who could at least potentially consent ($850 up to age 17), could be $50 or less in at least some contexts. But even if the comparison is between newborn and adult (age 18+) elective circumcisions, the difference could be as little as $700 if local anesthesia is used. For a point of comparison, the US national average cost of adult (elective) labiaplasty is $2,924 according to the American Society of Plastic Surgeons.[87] In either case, the cost of the adult procedure would be borne exclusively by the small minority of individuals who had an explicit, robust desire to undergo a genital-altering procedure, whereas those who did not have such a desire would pay nothing.

Morris *et al.* also argue that individuals circumcised as adults may need to take time off work. Any lost earnings that result from this would need to be taken into account as well. However, similar to the labiaplasty example, this financial loss would be moderated by the fact that it was voluntarily incurred to achieve a desired outcome, except in those rare cases where the procedure was medically necessary.[88] We discuss the ethical implications of these observations in a later section.

3. Ethical Issues

[86] See https://health.costhelper.com/circumcision.html (accessed November 3, 2019). We do not suggest that the estimates from this source are reliable or nationally representative; we use them purely for the sake of illustration. Please note that the most recent fees listed at the example clinic, Gentle Circumcision of California, are $900 for teens up to age 17 and $1600 for adults (18+). See http://www.gentlecircumcision.com/ (accessed November 3, 2019).

[87] See American Society of Plastic Surgeons (ASPS). (2019). Vaginal rejuvenation surgical options. ASPS. Available at https://www.plasticsurgery.org/cosmetic-procedures/vaginal-rejuvenation/labiaplasty (accessed November 3, 2019). The data cited are from 2018.

[88] Ungar-Sargon, *op cit*. note 73.

In the previous sections, we explored a range of empirical questions relating to the potential benefits and risks of circumcision before versus after an age of consent. We argued that the evidence for the medical benefits of circumcision is much less compelling than its strongest advocates claim. We also noted that only a few of these alleged benefits would apply to children below the age of consent. We then highlighted a number of problems with the arguments that neonatal circumcision carries fewer costs or disadvantages, whether general, medical, or financial. In the following section, however, we will simply grant for the sake of argument the claim of NNMC proponents that the greatest balance of *medical* benefits over risks applies to circumcisions performed in infancy, and especially the neonatal period, compared to circumcisions performed later in life. We argue that even if one grants this claim despite the contrary evidence surveyed above, it does not follow that NNMC is a morally acceptable practice.

A. Benefits

Circumcision Before Sexual Debut

The AAP was prompted to update its policy statement on NNMC in light of the three aforementioned RCTs in sub-Saharan Africa that reported a reduced risk of female-to-male HIV transmission following VMMC. Based on the inadequately supported assumption that this reported benefit can be meaningfully extrapolated to NNMC in the US, the AAP 2012 Task Force claimed that waiting until an age of consent to offer circumcision may be medically disadvantageous insofar as a certain percentage of US adolescents become sexually active before they can legally consent to sex.[89] But this claim needs to be put in context. Using data from the US Centers for Disease Control and Prevention (CDC) concerning female-to-male HIV transmission—and assuming for the sake of argument that the African data *could* be directly applied to the US—the researcher Sarah Bundick has argued as follows:

> If we assume that all 5,250 men who get HIV from a female sexual partner [per year] are not circumcised (though this is certainly not the case), the data suggest that about half of these infections — around 2,625 infections or ~5% of new infections — may have been prevented if the men had been circumcised. If we then factor in the number of men who are circumcised when they are infected (approximately 70-80% of American men are already circumcised), the number of infections that could have been prevented by

[89] AAP Task Force on Circumcision, "Cultural bias and circumcision," *op cit.* note 28, p. 803. The Task Force cites CDC data from 2011 showing that sexual debut had already occurred by the age of 13 in 6.2% of high school students.

> circumcision drops considerably. Taken together, the data suggest that the number of HIV infections that could be prevented in the US by promoting infant male circumcision is likely to be only in the hundreds per year — a tiny fraction of the estimated 50,000 new HIV infections.[90]

With respect to the specific claim of the AAP Task Force, it should be emphasized that the sub-population of males that is both (1) sexually active before a legal age of sexual consent and (2) at a meaningful risk of becoming infected with HIV (i.e., from either a similarly underage female sexual partner or an older female committing statutory rape), is much smaller than the sub-population analyzed by Bundick: a CDC study cited by the AAP Task Force shows that only 84 males ages 13-19 were infected with HIV via heterosexual contact in the entire US in 2010.[91] Indeed, the main driver of the HIV epidemic in the US is still homosexual sex between males (63% of all new infections),[92] and there is no consistent evidence that circumcision—at any age—prevents HIV in this subset of the population.[93] It is thus very different from the epidemic in sub-Saharan Africa.[94]

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[90] Bundick, S. (2009). Promoting male circumcision to reduce transmission of HIV: a flawed policy for the US. *Harvard Health and Human Rights Journal* (online). Available at https://www.hhrjournal.org/2009/08/promoting-infant-male-circumcision-to-reduce-transmission-of-hiv-a-flawed-policy-for-the-us/ (accessed November 3, 2019).

[91] Centers for Disease Control and Prevention. HIV surveillance in adolescents and young adults, p. 7. Available at https://web.archive.org/web/20120724001848/www.cdc.gov/hiv/topics/surveillance/resources/slides/adolescents/slides/Adolescents.pdf (accessed November 3, 2019). The AAP Task Force (*op cit.* note 28, p. 803) misleadingly cites the total number of HIV infections for this age group (2,266), which comprises data for both sexes and all vectors of infection.

[92] Centers for Disease Control and Prevention. HIV in the United States: at a glance. Available at http://www.cdc.gov/hiv/statistics/basics/ataglance.html (accessed November 3, 2019).

[93] A 2008 meta-analysis performed by the CDC and the five largest studies exploring whether there was a link between circumcision status and HIV prevalence or incidence in men engaging in homosexual activities failed to find an association that was statistically significant. Millett, G. A., Flores, S.A., Marks, G., Reed, J.B., & Herbst, J. H. (2008). Circumcision status and risk of HIV and sexually transmitted infections among men who have sex with men. *Journal of the American Medical Association*, 300, 1674-1684, (Errata *JAMA* 2009; 301, 1126-9); Solomon, S., Mehta, S., Srikrishnan, A., et al. (2015). High HIV prevalence and incidence among MSM across 12 cities in India. *AIDS*, 29, 723–731; Gust, D. A., Wiegand, R. E., Kretsinger, K., et al. (2010). Circumcision status and HIV infection among MSM: reanalysis of a Phase III HIV vaccine clinical trial. *AIDS*, 24, 1135-1143; Jozkowski, K., Rosenberger, J. G., Schick, V., Herbenick, D., Novak, D.S., & Reece, M. (2010). Relations between circumcision status, sexually transmitted infection history, and HIV serostatus among a national sample of men who have sex with men in the United States. *AIDS Patient Care & STD*, *24,* 465-470. Oster, A. M., Wiegand, R. E., Sionean, C., et al. (2011). Understanding disparities in HIV infection between black and white MSM in the United States. *AIDS*, *25*, 1103-1112. Subsequently, a 2019 meta-analysis calculated that circumcised men were at lower risk, but their findings are of questionable value considering the analysis also found evidence of between-study heterogeneity and significant publication bias. Yuan, T., Fitzpatrick, T., Ko, N.-Y., et al. (2019). Circumcision to prevent HIV and other sexually transmitted infections in men who have sex with men: a systematic review and meta-analysis of global data. *Lancet Global Health*, *7*, e436-47. When adjusted for these two factors, the difference was no longer seen: Van Howe, R.S. (2019). Is circumcision an answer for HIV prevention in men having sex with men? *Lancet Global Health*, 7(8):e1011.

[94] For reasons already discussed, this is not to say that NNMC would necessarily be justified in those settings (where the circumcision programs were originally framed as being voluntary); only that it may be particularly unjustified in the United States and epidemiologically similar countries.

Even granting, therefore, that a tiny fraction of the annual cases of female-to-male transmission of HIV in the US (and relevantly similar contexts) may occur in adolescents under the age of 15, likely engaged in unprotected sex, it is unclear how this could justify the non-voluntary removal of a psychosexually significant part of an individual's body long before sexual debut. Neonatally circumcised homosexual males—or heterosexual males who practice safe sex—will receive virtually no HIV-related benefit whilst bearing the burden of unsafe choices made by others. It is also worth remembering that circumcision is not a failsafe intervention: it does not prevent HIV (or any other sexually transmitted infection). Rather, it may merely reduce one's risk of contracting the virus by around half per coital act, assuming (for the sake of argument) that the African findings are both accurate and generalizable.

A more appropriate solution, then, would be to allow individuals to decide for themselves whether or not to undergo circumcision when they are old enough to understand what is at stake. If most decline to undergo the procedure even after being informed of its purported benefits, it is not safe to assume, as some circumcision proponents do, that this is only because they want to avoid the pain and disruption that accompany the procedure in later life.[95] Rather, these individuals may place a positive value on their foreskins or on the state of being genitally intact, or prefer alternative, less invasive, more effective means of preventing sexually transmitted infections (STIs). Consider that if adult men were forcibly circumcised to lower their risk of STIs, even on the paternalistic assumption that this was best *for them*, this would be seen as an impermissibly coercive public health strategy, amounting to state-sanctioned assault and battery. On the other hand, when infants—who are unable to offer any effective resistance—are forcibly circumcised, this is framed as a legitimate means of promoting personal and/or public health by advocates of NNMC.[96]

Let us turn finally to the question of the legal age of consent. NNMC proponents seem to assume that the age of consent for circumcision must be the same as the age of legal majority (18 years in most jurisdictions). By contrast, we argue that the age of consent for circumcision should be no earlier than the age of consent to sexual relations, which ranges from 16 to 18 across US states, often with close-in-age exemptions (so-called "Romeo and Juliet laws"). Such exemptions allow for even lower age limits when, for example, both partners are minors.[97] Alternatively, in discussing the age at which a child might be expected to provide their own fully informed consent

[95] Morris et al., *op cit.* note 13, p. 10.
[96] Morris, *op cit*. note 36.
[97] See https://www.ageofconsent.net/states (accessed November 3, 2019).

for certain medical or surgical interventions, the AAP's Committee on Bioethics suggested 14 years as a reasonable age.[98] Depending on the maturity, or perhaps the degree of sexual experience of the minor (if any), something approximating this age might well be appropriate as a lower limit for consenting to circumcision, consistent with the recent change in eligibility criteria for VMMC set by PEPFAR. The underlying point is simply that the person whose sexual or reproductive organ would be permanently affected by the procedure should have at least some reasonably well-informed say in whether or not they want to undergo the procedure. This is impossible in the neonatal period.

Cultural Benefits

Not all claimed benefits of circumcision before an age of consent relate to the medical aspects of the practice. Some have pointed to potential cultural or even spiritual benefits: in some cultures, for example, circumcision is commonly embraced as means of initiation that allows individuals to be regarded as full members of the group in which they are raised.[99] Such sociocultural benefits, it is proposed, are denied to those children who are not circumcised as early as possible: although they will be able to choose circumcision for themselves—later on—if that is what they desire, they will first have to go through childhood wondering why they look different to their peers, possibly being subject to ridicule in the proverbial locker room, and so on.

However, this argument is problematic, even if it is simply stipulated that the underlying empirical assumptions (about ridicule, etc.) are sound.[100] First, an argument of this form could be

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[98] Committee on Bioethics. (1995). Informed consent, parental permission, and assent in pediatric practice. *Pediatrics*, *95,* 314-317.

[99] Benatar & Benatar, *op cit.* note 22, pp. 43-45. See also Schlegel, A., & Barry III, H. (2017). Pain, fear, and circumcision in boys' adolescent initiation ceremonies. *Cross-Cultural Research*, 51(5), 435-463. Note that circumcision performed as a rite of passage into adulthood—i.e., around puberty—could in principle conform to our proposed consent threshold, i.e., if the child were mature enough to appreciate what was at stake in the initiation.

[100] It is not clear that such a stipulation would be justified. Advocates of NNMC do not typically cite any evidence that a child will in fact psychologically suffer if they end up being different from their peers in this particular respect. There is also a question about who the relevant peers are, depending on the context. In most Western countries where penile circumcision occurs, for example, it is usually limited to a minority sub-group, whereas most individuals in the larger society (of which the circumcised child is ostensibly also a part) are not circumcised. Assuming that the child is not entirely isolated from their non-circumcised peers, then, it could just as easily be predicted that they will be teased *for* being circumcised. In short, the teasing argument only works under an assumption of cultural homogeneity with high rates of circumcision; yet this is not the situation in most multi-cultural Western societies where the ethical debate about NNMC is primarily taking place. Even in the US state of Iowa, which has a relatively high rate of circumcision even for the US—and where the only empirical study on "locker room" teasing that we are aware of has been conducted—the data suggest that (1) penile size, not circumcision status, is the main source of teasing, and (2) even if a boy is teased for not being circumcised, this does not seem to cause much distress. In fact, "experiencing teasing or witnessing others being teased about penile appearance did not have an effect on the desire for a different penile appearance [and] being uncircumcised did not

applied to any cultural practice—no matter how unjust or harmful—upheld by social norms that stigmatize noncompliance. Second, whether NNMC really does confer a (net) cultural benefit depends on how the affected individual personally relates to the culture or sub-culture of their birth. Some may only identify with certain aspects of that culture; others may wish to reform it or leave it altogether. Even if one accepts (which seems unlikely in the modern world) that *most* individuals will not question the cultural commitments of their parents' generation, it is far from clear that the remainder will have been socially benefited by NNMC.[101]

B. Costs

Impact on sexual function

As discussed earlier, proponents of circumcision are generally dismissive of claims that removing the sensitive foreskin may affect sexual experience in a negative way or in a way that an individual might reasonably resent. For example, in a review paper cited frequently by these proponents,[102] Morris and Krieger rate studies purporting to show that circumcision has no effect on sexual pleasure (or even a positive effect) as being of higher quality, whilst they deem studies suggesting otherwise to be of lower quality.[103] But as Bossio *et al.* point out, the coding scheme employed by Morris and Krieger can be used in a questionable manner: for example, Morris and Krieger assigned the highest possible quality rating to a study of which one of them (Krieger) was the lead author,[104] despite the seemingly clear conflict of interest involved in assessing one's own

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increase the rate of personally experienced teasing." Alexander, S. E., Storm, D. W., & Cooper, C. S. (2015). Teasing in school locker rooms regarding penile appearance. *The Journal of Urology*, 193(3), 983-988, at 985.

[101] Ungar-Sargon, *op cit*. note 73; The Pew Forum on Religion & Public Life. (2008). *US religious landscape survey religious affiliation: diverse and dynamic*. Washington, DC: Pew Forum on Religion & Public Life.

[102] Morris, B. J., & Krieger, J.N. (2013). Does male circumcision affect sexual function, sensitivity, or satisfaction? — a systematic review. *Journal of Sexual Medicine,* 10(11), 2644-2657.

[103] Boyle, G. J. (2015). Does male circumcision adversely affect sexual sensation, function, or satisfaction? Critical comment on Morris and Krieger (2013). *Advances in Sexual Medicine*, 5(2), 7-12. The author reply is available at the journal website.

[104] Krieger, J. N., Mehta, S. D., Bailey, R. C., Agot, K., Ndinya-Achola, J. O., Parker, C., & Moses, S. (2008). Adult male circumcision: effects on sexual function and sexual satisfaction in Kisumu, Kenya. *Journal of Sexual Medicine*, 5, 2610–2622. According to Frisch (2011), "it should be recalled that a strong study design, such as a randomized controlled trial, does not offset the need for high-quality questionnaires. Having obtained the questionnaires from the authors … I am not surprised that these studies provided little evidence of a link between circumcision and various sexual difficulties. Several questions were too vague to capture possible differences between circumcised and not-yet circumcised participants (e.g. lack of a clear distinction between intercourse and masturbation-related sexual problems and no distinction between premature ejaculation and trouble or inability to reach orgasm). Thus, non-differential misclassification of sexual outcomes in these African trials probably favored the null hypothesis of no difference, whether an association was truly present or not." Frisch (2012). Author's Response to: Does sexual function survey in Denmark offer any support for male circumcision having an adverse effect? *International Journal of Epidemiology*, 41(1), 2012, 312–314.

work, while also failing to engage with or even mention the published criticisms of the study's materials.[105] Even more questionably, they assigned the highest possible quality rating for a study of its kind to a single paragraph in a popular book by Masters and Johnson, referring to an experiment purporting to show no difference in penile glans sensitivity between circumcised and non-circumcised individuals.[106] Yet the study was never subjected to peer review, and the paragraph provided no details of the tools, methodology, or statistical procedures used, when this information is absolutely required for properly evaluating claims of a "null effect."[107]

But even if one simply accepts the ratings assigned by Morris and Krieger, their own interpretation of the evidence does not support NNMC. This is because the studies they judge to be of the highest quality, which according to them "suggest that medical male circumcision has no adverse effect on sexual function, sensitivity, sexual sensation, or satisfaction," concern consensual, adult circumcision, not circumcision of newborns. If anything, then, the review by Morris and Krieger supports the performance of voluntary circumcision after an age of consent: for it is circumcision of this kind for which the evidence against adverse sexual outcomes is, by their own account, the strongest.

Of course, comprehensively assessing both the objective and subjective sexual outcomes of NNMC in a nuanced, scientific way is exceedingly difficult, and as such has never been done.[108] So the question might best be put in terms of the *burden of proof*. Arguably, this lies with those who propose to excise healthy tissue from the penis, especially when the person who would undergo this procedure is unable to provide their own consent.[109] We suggest that the arguments and evidence of NNMC advocates fall far short of meeting this burden. For one thing, proving a negative claim about adverse sexual outcomes is practically impossible: an absence of evidence

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[105] Krieger also appears to have a patent pending for a circumcision device of which he is the lead inventor: https://patents.google.com/patent/WO2013173804A1/ko (accessed November 3, 2019).

[106] Masters, W., & Johnson, V. (1966). *Human sexual response*. Boston, MA: Little Brown & Co, 189-91. The only study information provided is as follows: "A limited number of the male study-subject population was exposed to a brief clinical experiment designed to disprove the false premise of excessive sensitivity of the circumcised glans. The 35 uncircumcised males were matched at random with circumcised study subjects of similar ages. Routine neurologic testing for both exteroceptive and light tactile discrimination were conducted on the ventral and dorsal surfaces of the penile body, with particular attention directed toward the glans. No clinically significant difference could be established between the circumcised and the uncircumcised glans during these examinations."

[107] See Harms, C., & Lakens, D. (2018). Making 'null effects' informative: statistical techniques and inferential frameworks. *Journal of Clinical and Translational Research*, 3(Suppl 2), 382-393.

[108] Johnsdotter, S. (2013). Discourses on sexual pleasure after genital modifications: the fallacy of genital determinism (a response to J. Steven Svoboda). *Global Discourse,* 3(2), 256–265. See also Darby, R., & Cox, L. (2008). Objections of a sentimental character: the subjective dimensions of foreskin loss. *Matatu*, (37), 145-168.

[109] Adler, P. (March 30, 2016). Resolving circumcision controversies: the burden of proof and the benefit of the doubt. *Circumcision and the Law Blog* (blog). Available at https://circumcisionandthelawblog.wordpress.com/2016/03/30/resolving-circumcision-controversies-the-burden-of-proof-and-the-benefit-of-the-doubt/ (accessed November 3, 2019).

is not the same thing as evidence of absence unless one has accurately, robustly, and systematically measured all such possible outcomes with sufficient statistical power to detect them, should they exist.[110] Obviously this has not occurred. Moreover, there are many aspects of the foreskin that are not yet fully understood,[111] and increasing evidence that circumcision likely does alter subjective sexual experience in a variety of ways.[112] These alterations might not always be judged negatively: some men circumcised as adults, for instance, have indeed reported no difference or even an increase in sexual satisfaction, while other men have noted a clear decrease.[113] The point is that no one other than the owner of the penis is better positioned to determine whether they wish to assume the *risk* that circumcision might permanently compromise their sexual enjoyment. A person who seeks circumcision in adulthood also has the advantage of being able to explain to the surgeon exactly how they want the procedure performed. They are able to choose the "style" of cut that conforms to their own aesthetic preferences (e.g., "high and tight" or "low and loose"),[114] and because their penis is fully grown, the surgeon will be able to determine with precision its final or stable anatomy, vasculature and innervation.[115] In infancy, the penis is very small and the surgeon cannot know how big the organ will grow or exactly where to do the cutting in order to obtain the desired cosmetic result. Because this is essentially a guess, too much tissue is sometimes removed, as we noted earlier; or indeed too little tissue, such that re-operation may be pursued.[116]

Ignoring consent

NNMC proponents, both "strong" and "weak" (Box 2), argue that the requirement for consent should not apply to the person being circumcised, but instead to their parents. Parents are regularly required to make decisions for their children, this argument holds; why should circumcision not be one of them? As noted, we agree that parents should (or may) make many decisions on behalf of their children, but there are also certain decisions parents may *not* make on their behalf. The

[110] Boyle, G. J. (2018). Proving a negative? Methodological, statistical, and psychometric flaws in Ullmann et al. (2017) PTSD study. *Journal of Clinical and Translational Research*, 3(Suppl 2), 375-381.
[111] Martín-Alguacil et al., *op cit*. note 64.
[112] See for instance Sorrells et al., *op cit.* note 67., cited above. See also Frisch, M., Lindholm, M., & Grønbæk, M. (2011). Male circumcision and sexual function in men and women: a survey-based, cross-sectional study in Denmark. *International Journal of Epidemiology,* 40(5)*,* 1367–1381; Earp*, op cit.* note 67, and Martín-Alguacil et al., *op cit.* note 64.
[113] Kim, D., & Pang, M. G. (2007). The effect of male circumcision on sexuality. *BJUI,,* 99(3), 619-622.
[114] Savulescu, J. (2013). Male circumcision and the enhancement debate: harm reduction, not prohibition. *Journal of Medical Ethics,* 39, 416-417.
[115] Earp & Darby, *op cit*. note 71.
[116] Krill, A. J., Palmer, L. S., & Palmer, J. S. (2011). Complications of circumcision. *The Scientific World Journal*, 11, 2458–2468.

question is: to which category does the circumcision decision belong? One might attempt to answer this question by considering other kinds of bodily alterations that are already seen as off-limits in Western societies, and then triangulating between these cases and circumcision to see where the latter falls. Consider the example of tattooing. In many Western jurisdictions, it is illegal to tattoo a child's body, presumably because this permanently alters their body in a way the child may later reasonably regret or resent, before they are able to offer any meaningful consent.[117] In such jurisdictions, therefore, it would be illegal to tattoo a child's foreskin. Yet if a parent wanted to cut off the foreskin entirely, and then tattoo it, this would be permitted under current norms and standards. NNMC therefore seems anomalous when compared to other medically unnecessary child body modifications.

Against this view, it might be argued that NNMC, but not tattooing, carries at least some medical benefits (even if it is not clear that these benefits outweigh even the strictly medical risks, as previously discussed). This, then, could explain why the former is forbidden but the latter is not. So let us consider a different example, that of labiaplasty. Labiaplasty is similar to circumcision in that it removes genital tissue that is not necessary for sexual enjoyment but which nevertheless has certain properties—sensitivity to touch, elasticity, independent manipulability, and so forth—that allow for particular subjective sensations, during sex or foreplay, for example, that many people value positively. It is also similar to circumcision in that the genital tissue it removes is often warm and moist and may trap bacteria, can become infected or even cancerous, may be injured or torn during sexual activity, and requires regular washing to maintain good hygiene. Removing the labia, therefore, likely does confer at least some statistical health benefits in that it reduces the surface area of genital tissue that is not essential for sexual function (in some narrow sense) but which may on occasion pose a problem of one kind or another for its owner. In addition, it may plausibly confer at least some psychological or social benefits for some women, insofar as they prefer the aesthetics of a vulva that has been subjected to labiaplasty, or if it helps them avoid negative judgments from potential sexual partners (however inappropriate such judgments may be). Indeed, the World Health Organization (WHO) includes such broader psychosocial factors in its definition of health, so perhaps these potential outcomes should be counted as "health benefits" as well.

Now, let us assume for the sake of argument that performing labiaplasty in infancy is technically simpler, safer, and more cost effective—with a shorter healing time, and so on—than

[117] Chegwidden, J. (2009). *Response: Tasmanian Law Reform Institute Issues Paper No. 14: Non-therapeutic male circumcision,* 1-79.

labiaplasty performed on a consenting adult. Would any of these considerations, alone or together, make *non-voluntary* neonatal labiaplasty permissible, either morally or legally, in Western societies? It seems unlikely. As one of us has argued elsewhere, even if health benefits of some kind or degree do one day become reliably associated with neonatal labiaplasty, it is almost certain that opponents of the practice (which includes the WHO, the United Nations, all Western democracies, and the present authors) would continue to see clear moral problems:

> First, they would argue that healthy tissue is valuable in-and-of-itself, so should be counted in the "harm" column simply by virtue of being damaged or removed. Second, they would point to non-surgical means of preventing or treating infections, and suggest that these should be favored over more invasive methods. And third, they would bring up the language of rights: a girl has a right to grow up with her genitals intact, they would say, and decide for herself at an age of understanding whether she would like to have parts of them cut into or cut off.[118]

The same arguments, we suggest, apply to NNMC.

Irreversibility

What about non-physical ways of shaping of a child? Eldar Sarajlic has contrasted circumcision with religious indoctrination. He notes that while parents are permitted to raise their children within a particular religion, there is nothing inherently irreversible about religious indoctrination (he argues, however, that religious indoctrination is only acceptable if certain efforts are made to expose children to other ways of thinking outside the home).[119] In any event, a great many people raised within a particular religious or cultural tradition later abandon certain associated beliefs and practices.[120] With circumcision, however, no such abandonment is possible. As an analogy, imagine that Christian parents wanted to tattoo "a Christian cross on their child's body; the fact that the child can later distance himself from Christianity does not make the tattoo legitimate and we could understand his upset about having to carry this religiously imposed, permanent mark, which [he] might understandably perceive as overstepping a boundary (and therefore as abuse)."[121]

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[118] Earp, B. D. (2017). Does female genital mutilation have health benefits? The problem with medicalizing morality. *Journal of Medical Ethics Blog*. Available at https://blogs.bmj.com/medical-ethics/2017/08/15/does-female-genital-mutilation-have-health-benefits-the-problem-with-medicalizing-morality/ (accessed November 3, 2019).
[119] Sarajlic, E. (2014). Can culture justify infant circumcision? *Res Publica,* 20(4), 327-343.
[120] The Pew Forum, *op cit*. note 101.
[121] Möller, K. (2017). Ritual male circumcision and parental authority. *Jurisprudence*, 8(3), 461-479, p. 472.

This irreversibility is a cost that must be taken seriously. Some NNMC proponents compare this with the cost that a genitally intact person must bear if they decide to undergo circumcision later in life,[122] but a fundamental asymmetry remains. Any person who was not circumcised, but wishes to be, can undertake the surgery voluntarily. A person who was circumcised without their consent and who resents what was done to their body, by contrast, has no ethically comparable recourse.[123]

It is true, as discussed earlier, that choosing circumcision in later life is not cost-free: one must take time off school or work and refrain from sexual activity for a number of weeks. However, older children and adults currently seem willing to take time off work for other valued projects (e.g., going on a vacation) or to refrain from sexual activity after other elective surgeries (e.g., a vasectomy or a labiaplasty). It is unclear why circumcision should be treated as a special kind of burden. Moreover, NNMC proponents do not account for comparable disruptions in the neonatal period. The developmental needs of an infant are very different to those of an adult. However, since the infant cannot report on these needs, there is an epistemic asymmetry. Adults are in a position of being much more aware of—or more easily able to imagine—the kinds of disruptions they would incur. Such disruptions are therefore easier to identify and perhaps to measure. But this does not entail that disruptions for the infant are any less significant; there is simply not enough quality research into the question.

Vaccination analogy

NNMC proponents often liken the procedure to a vaccination, which as Morris *et al.* point out, is "similarly performed before the child is old enough to consent and which carr[ies] similar risks of complications."[124] However, even if the likelihood of complications is similar between the two procedures (we are not suggesting this is actually the case), the *kind* of complications will often be dissimilar—for example, an allergic skin reaction versus a genital laceration—and different people may assign different weights to such variable complications. Moreover, in the case of

[122] Benatar & Benatar, *op cit*. note 22, p. 37.
[123] They may attempt one the various techniques of so-called foreskin restoration (whereby weights, tapes, or other devices are attached to the remaining penile shaft skin over several months or years to attempt to stretch it forward) but this is really misnomer. The foreskin is not actually restored; nor are the specialized nerve endings that were excised with it. Rather, a pseudo-foreskin is at best created, which some individuals evidently judge to be better than no foreskin at all. See Schultheiss, D., Truss, M. C., Stief, C. G., & Jonas, U. (1998). Uncircumcision: a historical review of preputial restoration. *Plastic and Reconstructive Surgery*, 101(7), 1990-1998. For evidence from sales records that more than 500,000 men in the English-speaking world are likely engaged such burdensome foreskin "restoration," see Earp, B. D. (2016). Between moral relativism and moral hypocrisy: reframing the debate on "FGM." *Kennedy Institute of Ethics Journal*, 26(2), 105-144 (page E-9 of the supplementary material).
[124] Morris et al., *op cit*. note 13.

circumcision, the risk is concentrated on a particular part of the body—namely, the penis—that is widely seen as having a special kind of value or significance compared to many other body parts, including the typical sites of vaccine administration such as the arm or thigh.

There are other differences as well. As McIntyre *et al.* note, "vaccination satisfies ethical criteria for preventive interventions in children: it is effective, minimally invasive, and associated with significant societal benefits."[125] We take "effectiveness" here to refer to an improvement to the individual's health. As already argued, the prevailing opinion of international medical authorities is that the potential, statistical health benefits of NNMC, especially for individuals in developed countries, fail to outweigh the risks and harms of the procedure. Circumcision does not prevent any diseases or maladies; at most, it may reduce the (typically already low) absolute risk of these, and it is rarely the least harmful or the most effective available means of reducing this risk. This stands in stark contrast to the measles vaccine, for example, which immunizes 95% of those who receive it against a disease with notable morbidity that can spread quickly through incidental, unknowing contact with infected air particles in public spaces.[126] (Compare this to the diseases that circumcision is said to protect against, which are primarily spread slowly through intimate, knowing contact in private spaces). Given that circumcision involves cutting and removing healthy tissue from a sensitive, sex-specific organ—versus a needle-prick that does not remove tissue from any part of the body, nor even typically leave a very salient lasting mark—it does not pass the minimally invasive test for prophylactic interventions. Indeed, imagine a vaccine that could only be administered to non-consenting girls and which required the removal of a sizable portion of the healthy tissue from their vulvas in order for it to work. Such a vaccine would rightly not be allowed. As Somerville points out, it is not enough to show that NNMC can decrease one's likelihood of contracting various diseases: because it is a non-voluntary surgical procedure, it must also be "the only reasonable way to obtain these benefits."[127] If there are less

[125] McIntyre, P., Williams, A., & Leask, J. (2003). Refusal of parents to vaccinate: dereliction of duty or legitimate personal choice? *The Medical Journal of Australia*, 178(4), 150, citing Hodges, F. M., Svoboda, J. S., & Van Howe, R. S. (2002). Prophylactic interventions on children: balancing human rights with public health. *Journal of Medical Ethics*, 28(1), 10-16.

[126] See Centers for Disease Control and Prevention (CDC). Measles cases and outbreaks. CDC.gov. Available at http://www.cdc.gov/measles/cases-outbreaks.html (accessed November 3, 2019). Or consider the flu vaccine. While it is not as effective from year to year as the measles's vaccine (according to the CDC, it "reduces the risk of flu illness by between 40% and 60% among the overall population during seasons when most circulating flu viruses are well-matched to the flu vaccine"), it is the best available option for reducing the risk of a highly contagious illness that simultaneously satisfies the minimally invasive criterion. See CDC. Vaccine effectiveness: How well do the flu vaccines work? CDC.gov. Available at https://www.cdc.gov/flu/vaccines-work/vaccineeffect.htm (accessed November 3, 2019).

[127] Somerville, M. (2000). *The ethical canary: science, society and the human spirit*. Toronto: Viking, p. 205.

invasive means of obtaining the benefits, as indeed there are (e.g., basic hygiene and safe sex practices, or indeed actual vaccinations), these should be preferred.

Benatar and Benatar argue that just because one intervention is more invasive than another, it does not follow that it is excessively invasive.[128] They note, for instance, that oral vaccinations are not as invasive as those by injection, which are in turn not as invasive as circumcision. Notice, however, that neither of the first two interventions, though somewhat invasive, irreversibly alters one's gross anatomy and external physical appearance—much less a physically and symbolically salient, psychosexually significant aspect thereof.[129] The immediate side effects and any long-term complications of a vaccination may be regretted, but in the absence of these, it difficult to explain how one could reasonably come to regret the vaccination *per se*. By contrast, hundreds of thousands of individuals seem to resent the non-voluntary removal of their foreskins even in the absence of clinically significant complications.[130]

As for the last condition of significant societal benefits, based on our earlier discussion, this too would not be met by NNMC in developed countries (and the jury is still out with respect to sub-Saharan Africa). Lyons has observed that a number of "parents are already skeptical about the risk-benefit ratio of conventional vaccines." The "possible misrepresentation of a surgical procedure [especially one as controversial as NNMC] as an effective vaccine seems unlikely to improve trust."[131] In fact, not only might this fail to encourage more people to circumcise their infants, he argues; it could lead to fewer parents vaccinating their children.[132]

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[128] Benatar, D., & Benatar, M. (2003). How not to argue about circumcision. *American Journal of Bioethics*, 3(2), W1-W9.

[129] Earp, B. D., & Steinfeld, R. (2018). Genital autonomy and sexual well-being. *Current Sexual Health Reports*, 10(1), 7-17.

[130] See Earp, "Between moral relativism," *op cit.* note 123. The figure reported there (at least 500,000) is the best available, conservative estimate of the number of males in the English-speaking world actively engaged in "foreskin restoration." As this is a very burdensome and time-consuming process that, if successful, creates a pseudo-prepuce only, it stands to reason that these individuals are/were highly resentful of having had their foreskins removed. More recently, a demographically diverse (albeit not nationally representative) survey of 999 US American men—carefully designed to avoid oversampling of individuals who might have concerns about circumcision—found that of the 771 respondents who reported being circumcised, 105 (13.6%) reported wishing they were the opposite circumcision status, with a quarter of that sub-group stating that they would "seriously consider" changing their circumcision status if it were possible (i.e., through a foreskin restoration process). Please note that a similar percentage of genitally intact individuals (15.9%) reported wishing they were the opposite circumcision status; however, as we argue, such individuals still have an option open to them, whereas those who reported resentment about being circumcised have no comparable recourse. See Earp, B. D., Sardi, L., & Jellison, W. (2018). False beliefs predict increased circumcision satisfaction in a sample of US American men. *Culture, Health & Sexuality, 20(8)*, 945–959. For further evidence of resentment among circumcised individuals and related discussions, see Hammond & Carmack and Bossio & Pukall, *op cit.* note 72.

[131] Lyons, B. (2013). Male infant circumcision as a 'HIV vaccine'. *Public Health Ethics,* 6(1), 90-103, 99.

[132] *Ibid.*

Medical complications

As noted previously, assuming that the figures provided by Morris *et al.* about the medical complications of NNMC versus circumcision later in life are correct (contrary to the recent evidence from PEPFAR), the relative risk does not appear to be much greater, if it is greater at all. And according to NNMC advocates, the *absolute* rate of complications in both cases is "low," with "virtually all" being mild and treatable if and when they do occur.[133] But if the medical risks in both cases are "low" in absolute terms and otherwise mild or treatable, as NNMC advocates maintain, then the relative difference between the risks should not be considered a morally weighty factor.[134] In the case of an adult, any complications that do occur are the result of a procedure that was voluntarily undertaken, after the adult had been informed about the nature and likelihood of these very risks. This is not possible in the case of the neonate or young child. Generally speaking, risks imposed on a person who is aware of the risks and agrees to assume them in light of their own considered preferences and values are much less morally objectionable than risks imposed on a person who is not aware of the risks and/or does not agree to assume them.

Financial costs

The claim that NNMC is generally less expensive than adult circumcision is plausible and we do not wish to dispute this. However, the cost for individuals between ages 14 and 17 is in some cases far less than for those aged 18 and older (see above discussion), and we have suggested that these individuals, in contrast to infants, may at least in principle be able to provide their own informed consent. As stated earlier, one of the main reasons adult circumcision tends to cost so much is that it is usually performed under general anesthesia, which is more effective than local anesthesia in preventing the pain associated with the procedure. Neonates receive either local anesthesia (which could include something as mild as a topical cream) or none at all, and are in no position to communicate their pain levels explicitly. Even among NNMC defenders, it is usually accepted that infants experience significant pain both during and after the procedure.[135] This remains the case under local anesthesia, because even the most effective form, dorsal-penile nerve block, is itself painful to administer and sometimes fails.[136] From an ethical perspective,

[133] Morris, B. J., & Green, E. C. (2014). Circumcision, male. *Blackwell Encyclopedia of Health, Illness, Behavior, and Society* (pp. 253-6). Hoboken: Wiley-Blackwell.
[134] Carmack, A., Notini, L., & Earp, B. D. (2016). Should surgery for hypospadias be performed before an age of consent? *The Journal of Sex Research*, 53(8), 1047–1058.
[135] See Benatar & Benatar, *op cit*. note 22.
[136] See Frisch & Earp, *op cit.* note 21, for references and discussion.

using less effective anesthesia on a more vulnerable individual—who cannot refuse the procedure—in order to drive down costs should be called into question.

Finally, the cost comparison between neonatal and adult circumcision is in one sense misleading. One must also consider the cost comparison between either neonatal or adult circumcision, on the one hand, and *no* circumcision on the other. The latter option has no procedural costs tied to it, and it appears to be the default preference of the vast majority of genitally intact individuals. If a particular individual does choose to be circumcised absent medical necessity in adulthood, this will likely cost more to them in strictly financial terms than it would have cost their parents if the same individual had been circumcised in the neonatal period. But in this case, the (incalculable) moral costs of NNMC will have been avoided.

4. Conclusion

In this paper, we have interrogated the common view that NNMC carries fewer risks and confers greater benefits than circumcision later in life. We have shown that NNMC proponents who advance this view are usually thinking only in terms of a narrow analysis of the medical benefits (disease prevention) and risks (surgical complications). When one considers potential benefits and risks in a more comprehensive manner, including in light of more recent evidence, this view is less compelling.

Circumcision removes healthy genital tissue that is *prima facie* valuable, while imposing the risk of a number of surgical complications, ranging from mild to severe, the full extent of which is understudied and unknown.[137] The favorable attitude of NNMC proponents toward the procedure appears to be at odds with that taken towards medically unnecessary surgeries more generally, especially when the affected person cannot consent.[138] As a California Appeals Court stated in 2006, it "seems self-evident that unnecessary surgery is injurious and causes harm to a patient. Even if a surgery is executed flawlessly, if the surgery were unnecessary, *the surgery in and of itself constitutes harm*."[139]

There are a number of reasons to prefer circumcision after an age of consent, as we have discussed. The most important, however, is that it preserves freedom of choice. Those who were

[137] AAP Task Force on Circumcision, *op cit*. note 29, p. e772.
[138] This is beginning to be recognised, however. See the Case No. LJ13C00295: Re B and G (Children) (No 2) [2015] EWFC 3, p. 69, in which Judge James Munby ruled that 'circumcision' of male, as well as female, children "amounts to significant harm." Available at https://www.familylaw.co.uk/news_and_comment/re-b-and-g-children-no-2-2015-ewfc-3 (accessed November 3, 2019).
[139] Tortorella v. Castro, 43 Cal. Rptr. 3d 853 - Cal: Court of Appeal, 2nd Appellate Dist., 3rd Div. 2006, emphasis added.

not circumcised in childhood, but wish they had been, still have the option of circumcision available to them; whereas those who were circumcised in childhood, but wish they had not been, cannot undo what has already happened. In our view, the concerns of those who have had a healthy part of their intimate anatomy removed, resent this, and have no adequate recourse, should be taken more seriously than the concerns of those who did not have a healthy part of their intimate anatomy removed, resent this, and still have a recourse. Accordingly, from a moral point of view—which includes a medical cost-benefit analysis as only one part of the overall ethical analysis—we find that medically unnecessary circumcision should not be performed on minors who cannot provide ethically valid consent to the procedure.

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